Bayesian methods for uncertainty quantification

Part II – Monte Carlo statistical methods

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Outline

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Bayesian inference

Bayesian model

- Probabilistic modeling of the observations/data: $\mathbf{y}|\mathbf{x} \sim f(\mathbf{y}|\mathbf{x})$ (likelihood function)
- Probabilistic modeling of the prior knowledge: $\mathbf{x} \sim f(\mathbf{x})$ (prior distribution)
- Posterior distribution defined as

$$f(\mathbf{x}|\mathbf{y}) \propto f(\mathbf{y}|\mathbf{x})f(\mathbf{x})$$

MAP estimation

Maximizing the (log-)posterior distribution yields

$$\hat{\mathbf{x}}_{\mathrm{MAP}} = \underset{\mathbf{x}}{\operatorname{argmin}} \underbrace{-\log f(\mathbf{y}|\mathbf{x})}_{\text{data-fitting}} \underbrace{-\log f(\mathbf{x})}_{\text{regularization}}$$

which gives, in case of $\ell_2 - \ell_2$ problem (ridge regression)

$$\hat{\mathbf{x}}_{\mathrm{MAP}} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 + \lambda \|\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}}\|_2^2$$

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Bayesian inference as a variational problem

- Likelihood function rewritten as

$$f(\mathbf{y}|\mathbf{x}) \propto \exp\left(-\mathcal{D}(\mathbf{y}|\mathcal{M}(\mathbf{x}))\right)$$

  where $\mathcal{D}(\cdot|\cdot)$ is a discrepancy measure

- Prior distribution rewritten as

$$f(\mathbf{x}) \propto \exp\left(-\lambda\mathcal{R}(\mathbf{x})\right)$$

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Given $\mathbf{y}$ and $\mathbf{H}$, one are looking for $\hat{\mathbf{x}} \in \mathbb{R}^M$ which maximizes the neg-log-posterior

$$\hat{\mathbf{x}} = \underset{\mathbf{x}\in\mathbb{R}^M}{\operatorname{argmin}} \underbrace{\mathcal{D}(\mathbf{y}|\mathcal{M}(\mathbf{x})) + \lambda\mathcal{R}(\mathbf{x})}_{\mathcal{J}_{\mathbf{y}}(\mathbf{x})}$$

$\rightarrow$ an optimization problem

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$\rightarrow$ an optimization problem

Bayesian inference

MAP estimation... as an optimization problem

$$\hat{\mathbf{x}} \in \underset{\mathbf{x} \in \mathcal{X}}{\operatorname{argmin}} \, \mathcal{J}(\mathbf{x})$$

No closed-form solution (or too difficult to compute)

→ resorting to optimization algorithms!

→ approximating the solution thanks to an iterative solution
⇝ *for instance, a gradient-descent algorithm*

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha^{(t)} \nabla \mathcal{J}_{\mathbf{y}}(\mathbf{x}^{(t)})$$

However...

Bayesian inference

MAP estimation... as an optimization problem

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Bayesian inference
MAP estimation... as an optimization problem

Challenges

- large scale problem: $\mathbf{x} \in \mathbb{R}^M$ with $M \gg 1$
  $\rightsquigarrow$ *solution space of large dimension*
- "big data" context: $\mathbf{y} \in \mathbb{R}^N$ with $N \gg 1$
  $\rightsquigarrow$ *difficulté à évaluer globalement $\mathcal{J}_{\mathbf{y}}(\cdot)$*
- distributed or online processing: $\mathbf{y} = \cup_{j=1}^{K} \{\mathbf{y}_j\}$
  $\rightsquigarrow$ *data shared among several cores/nodes/workers*
  $\rightsquigarrow$ *online data acquisition*
- space of the constraints $\mathcal{X}$ non trivial
  $\rightsquigarrow$ *how to satisfy these constraints?*
- non-smooth criterion: $\nabla \mathcal{J}_{\mathbf{y}}(\cdot)$ undefined
  $\rightsquigarrow$ *how to getting rid of $\nabla \mathcal{J}_{\mathbf{y}}(\cdot)$?*
- non-convex problem
  $\rightsquigarrow$ *existence of local minima (at best)*
  $\rightsquigarrow$ *existence of several solutions (at worst)*
- ...

Bayesian inference

MAP estimation... as an optimization problem

Some opportunities

- large scale problem: $\mathbf{x} \in \mathbb{R}^M$ with $M \gg 1$
  $\rightsquigarrow$ *large scale optimization*
- "big data" context: $\mathbf{y} \in \mathbb{R}^N$ with $N \gg 1$
  $\rightsquigarrow$ *stochastic optimization*
- distributed or online processing: $\mathbf{y} = \cup_{j=1}^{K} \{\mathbf{y}_j\}$
  $\rightsquigarrow$ *distributed or federated optimization*
  $\rightsquigarrow$ *online or adaptive optimization*
- space of the constraints $\mathcal{X}$ non trivial
  $\rightsquigarrow$ *constrained optimization*
- non-smooth criterion: $\nabla \mathcal{J}_{\mathbf{y}}(\cdot)$ undefined
  $\rightsquigarrow$ *derivative-free or proximal optimization*
- non-convex problem
  $\rightsquigarrow$ *robust optimization (worst-case)*
  $\rightsquigarrow$ *global optimization*
- ...

Bayesian inference
Beyond MAP estimation

MMSE estimation
Computing the posterior mean (or MMSE) estimator boils down to compute

$$\hat{\mathbf{x}}_{\mathrm{MMSE}} = \mathrm{E}\left[\mathbf{x}|\mathbf{y}\right] = \int \mathbf{x} f(\mathbf{x}|\mathbf{y}) d\mathbf{x}$$

Credible intervals
Deriving the credible interval boils down to compute integrals of the form

$$\mathrm{P}[\mathbf{x} \in \mathbb{S}_\alpha | \mathbf{y}] = \int_{\mathbb{S}_\alpha} f(\mathbf{x}|\mathbf{y}) d\mathbf{x} = \int \mathbf{1}_{\mathbb{S}_\alpha}(\mathbf{x}) f(\mathbf{x}|\mathbf{y}) d\mathbf{x}$$

Bayesian "quantities"
More generally, one aims at evaluating

$$\mathrm{E}\left[G(\mathbf{x})\right] = \int G(\mathbf{x}) f(\mathbf{x}|\mathbf{y}) d\mathbf{x}$$

where $G(\cdot)$ is a given function and $f(\mathbf{x}|\mathbf{y})$ is the posterior distribution.
$\rightarrow$ an integration problem

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Challenges

- the parameter **x** belongs to a large dimensional space
- the function $G(\cdot)$ is not trivial
- the posterior distribution $f\left(\mathbf{x}|\mathbf{y}\right)$ departs from a “standard” distribution...
- ... or is even not defined by a closed-form expression since

$$f(\mathbf{x}|\mathbf{y}) \propto f(\mathbf{y}|\mathbf{x}) f(\mathbf{x})$$

where the normalizing constant (evidence) $f(\mathbf{y})$ is difficult to evaluate

$$f(\mathbf{y}) = \int f(\mathbf{y}|\mathbf{x}) f(\mathbf{x}) d\mathbf{x}$$

Alternative: Monte Carlo integration

- Generating samples $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(T)}$ distributed according to $f(\mathbf{x}|\mathbf{y})$
- approximating $\mathrm{E}\left[G(\mathbf{x})\right]$ by

$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} G\left(\mathbf{x}^{(t)}\right)$$

“Proof”: (strong) law of large numbers...

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Problem statement

For a given random variable[1] $\theta \sim f(\theta)$, one seeks to evaluate

$$\mathrm{E}\left[G(\theta)\right] = \int G(\theta) f(\theta) d\theta$$

Application to Bayesian inference
Adopting the previous notations

$$\begin{aligned} \theta &= \mathbf{x} \\ G(\theta) &= \mathbf{x} \\ f(\theta) &= f(\mathbf{x}|\mathbf{y}) \end{aligned}$$

then

$$\begin{aligned} \mathrm{E}\left[G(\theta)\right] &= \mathrm{E}\left[\mathbf{x}|\mathbf{y}\right] \\ &= \int \mathbf{x} f(\mathbf{x}|\mathbf{y}) d\mathbf{x} \\ &= \hat{\mathbf{x}}_{\mathrm{MMSE}} \end{aligned}$$

[1] One adopts a generic notations where $f(\theta)$ may denote a posterior distribution.

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Monte Carlo integration

Solution

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$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} G\left(\theta^{(t)}\right)$$

Properties

Asymptotically, one has (in a certain sense that should be specified...)

$$\bar{G}_T \sim \mathcal{N}\left(\mathrm{E}\left[G(\theta)\right], \nu_T^2\right)$$

$\rightarrow$ also provides confidence intervals...

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Monte Carlo methods

Denomination

- also referred to as "**stochastic simulation**"
- name inspired by the eponymous casino in Monaco

Definition

- a class of algorithms
- rely on the (pseudo-)random generation of numbers by a computer
- these random draws are used to approach the numerical value of a quantity of interest

and it is not 'recent' (WWII context).

Example: approximation of π

Problem statement

Let consider a 2-dimensional random variable $\boldsymbol{\theta} = [\theta_1, \theta_2]^\mathsf{T}$ such that θ_1 and θ_2 are independent and identically distributed (i.i.d.) with

- $\theta_1 \sim \mathcal{U}(0,1)$
- $\theta_2 \sim \mathcal{U}(0,1)$

One shows that

$$\mathrm{P}\left[\|\boldsymbol{\theta}\|_2^2 \leq 1\right] = 4 \times \mathrm{P}\left[\boldsymbol{\theta} \in \mathcal{D}^+(0,1)\right] = \pi$$

thus

$$\pi = 4 \times \mathrm{P}\left[\boldsymbol{\theta} \in \mathcal{D}^+(0,1)\right] = 4 \times \int_{\mathcal{D}^+(0,1)} f(\boldsymbol{\theta}) d\boldsymbol{\theta} = 4 \times \int_{(0,1)^2} \mathbf{1}_{\mathcal{D}^+(0,1)}(\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

Monte Carlo integration

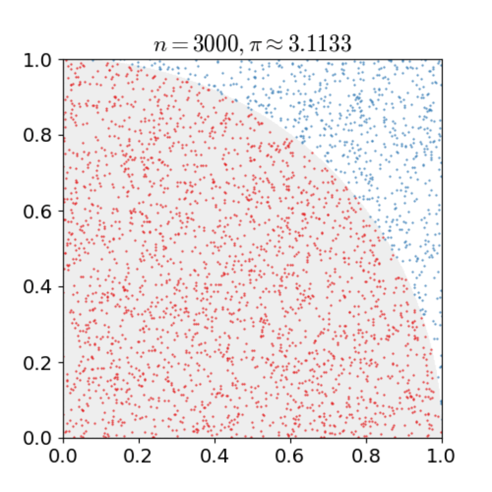
For $t = 1, \ldots, T$

- Draw $\theta_1^{(t)} \sim \mathcal{U}(0,1)$
- Draw $\theta_2^{(t)} \sim \mathcal{U}(0,1)$

and then approximate π by

$$\bar{\pi} = 4 \times \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}_{\mathcal{D}^+(0,1)}\left(\boldsymbol{\theta}^{(t)}\right) = 4 \times \frac{1}{T} \mathrm{card}\left\{\boldsymbol{\theta}^{(t)} \mid \left\|\boldsymbol{\theta}^{(t)}\right\|_2^2 \leq 1\right\}$$

Example: approximation of π



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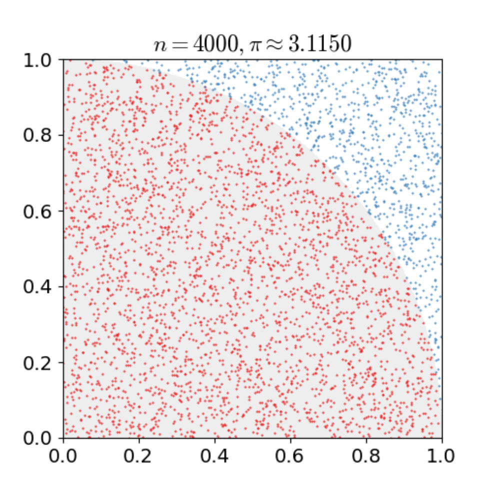

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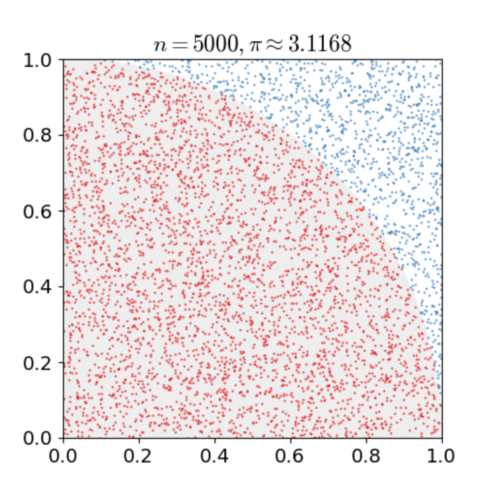

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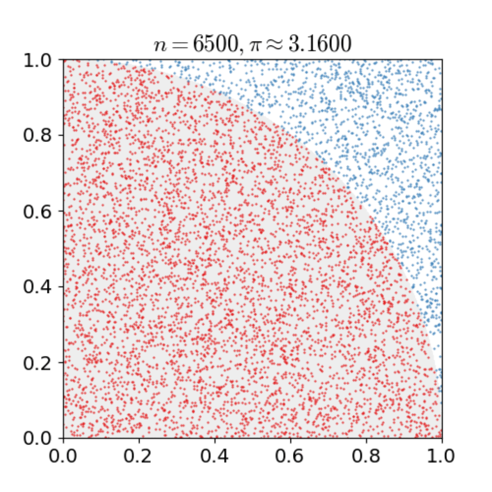

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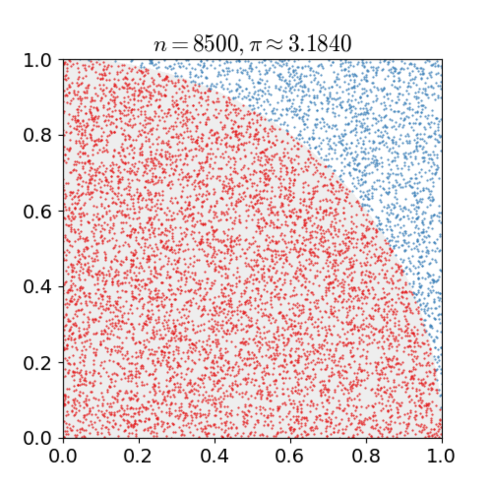

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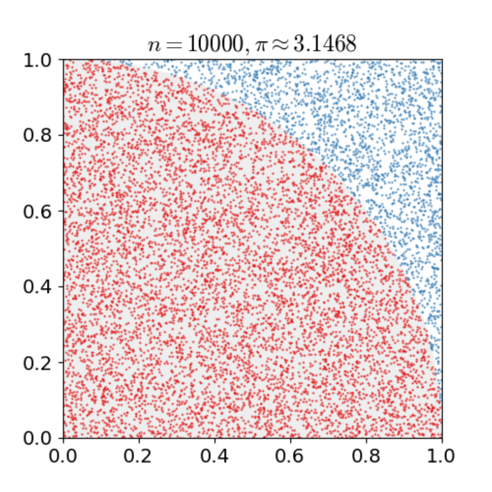

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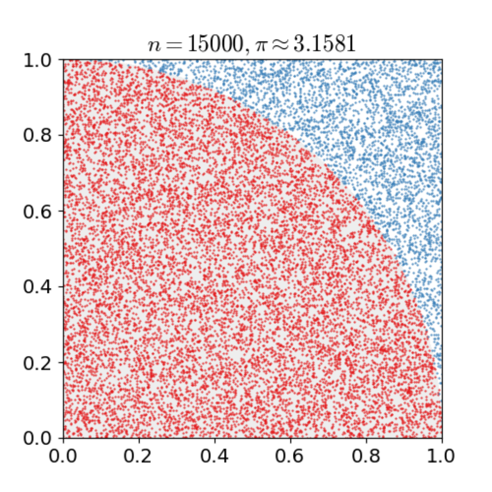

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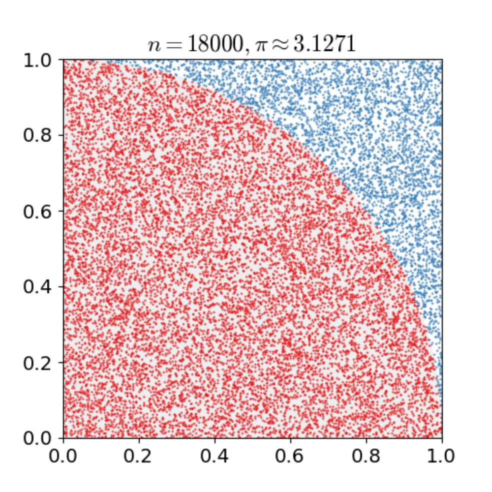

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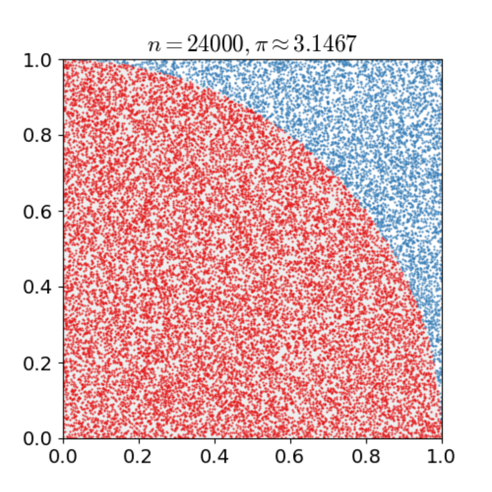

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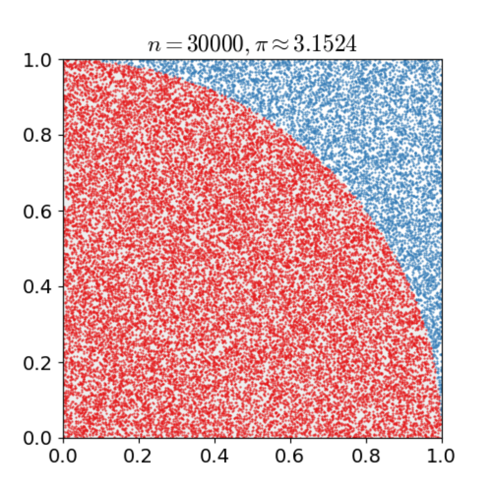
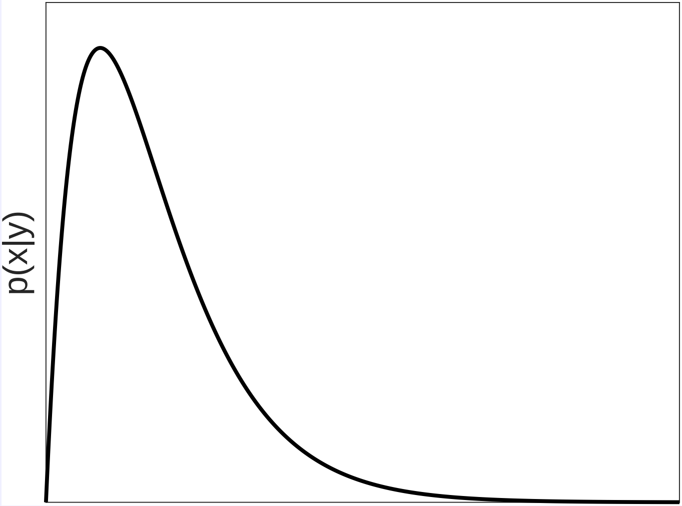
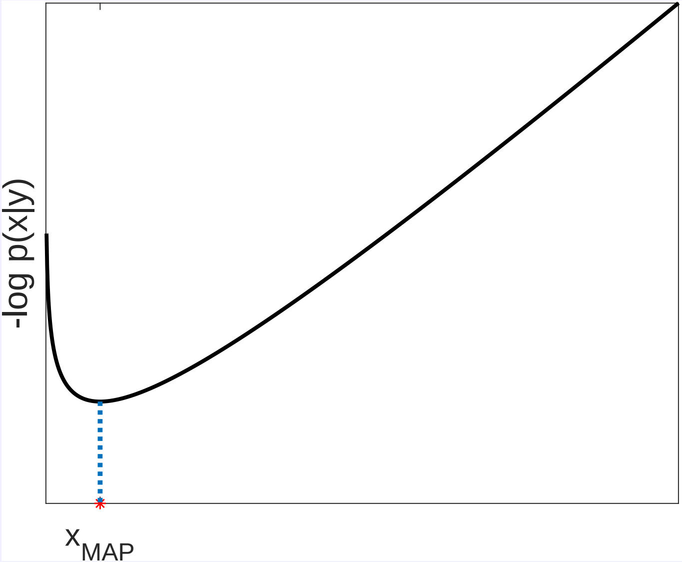


Monte Carlo methods for Bayesian inference
Computing the posterior distribution



Monte Carlo methods for Bayesian inference
MAP estimation



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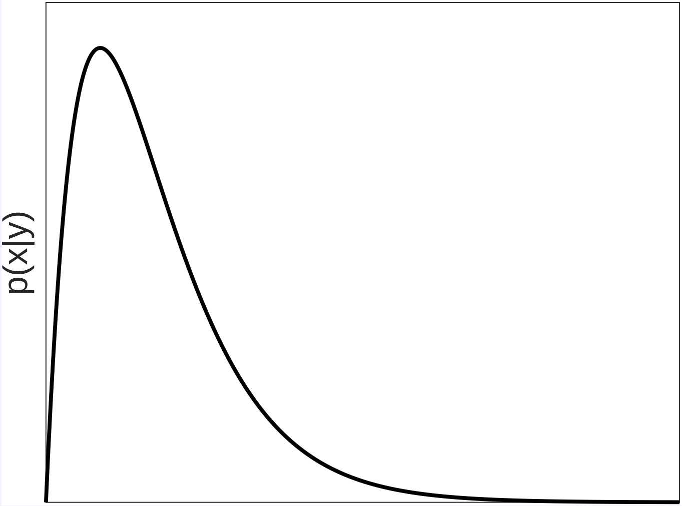
Sampling from the posterior distribution



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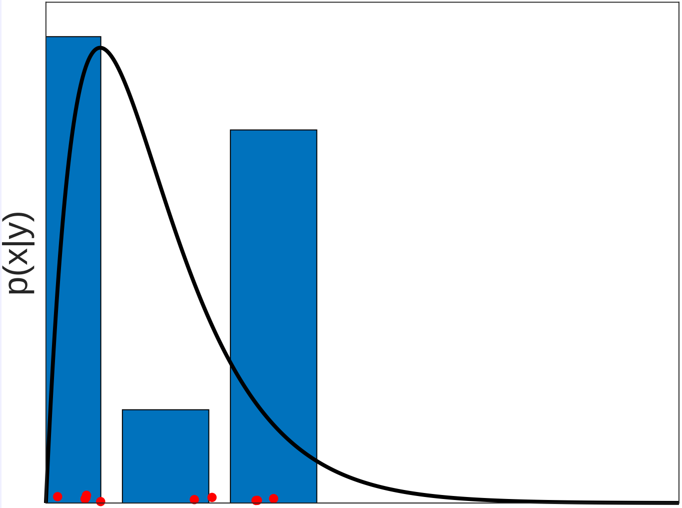

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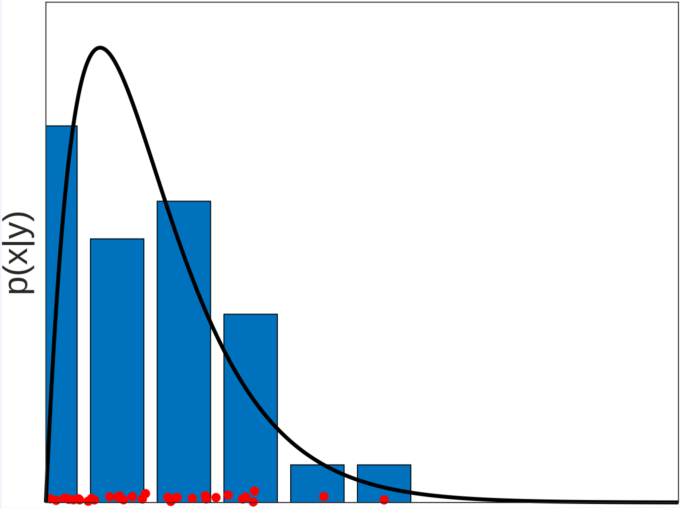

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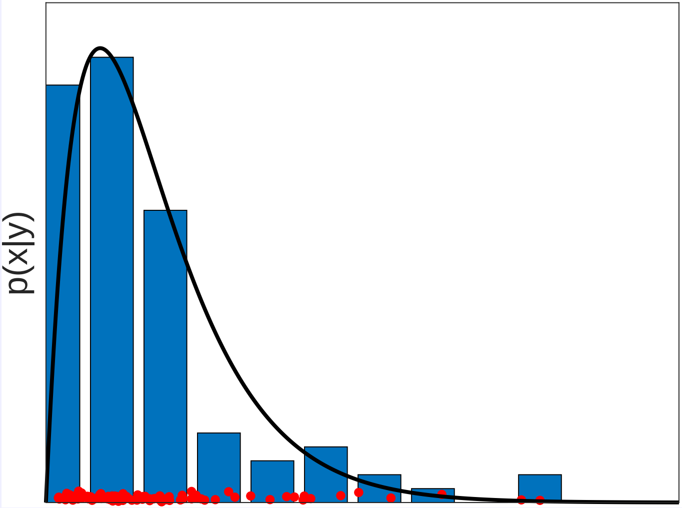

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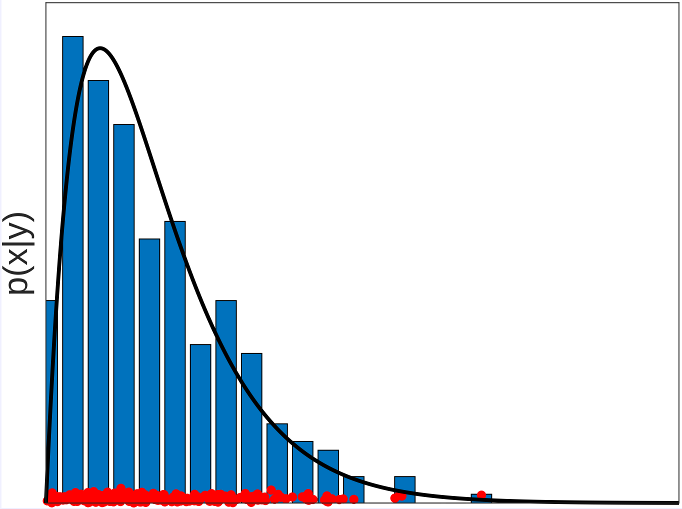

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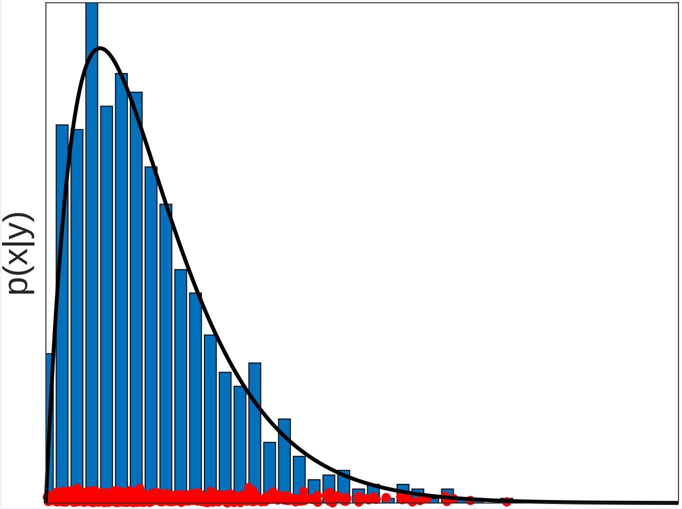

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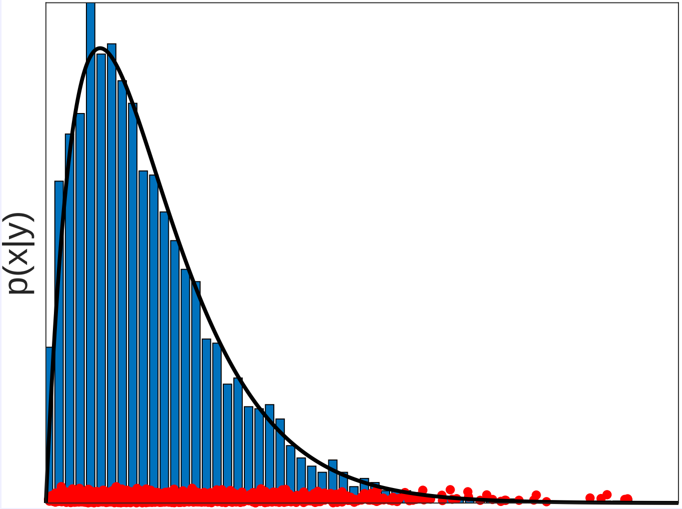

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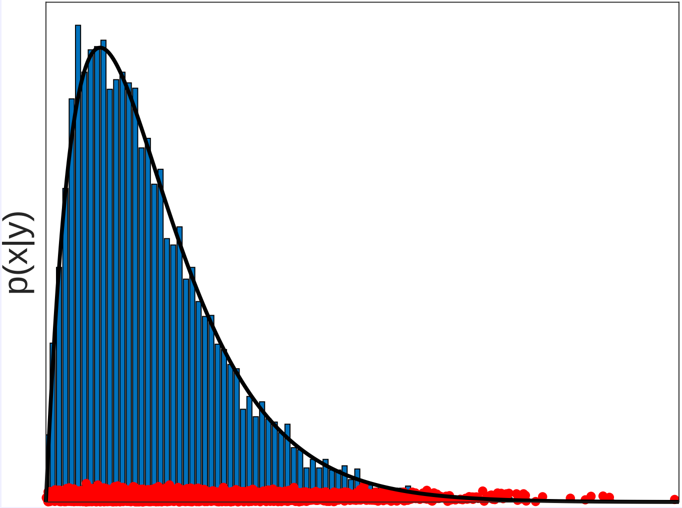

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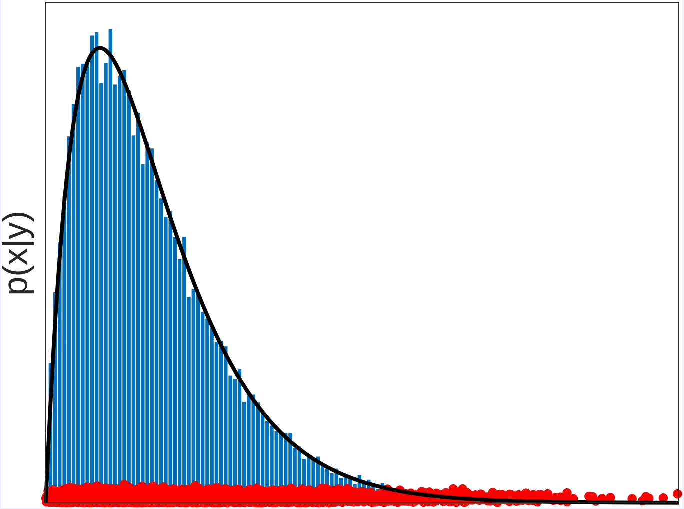
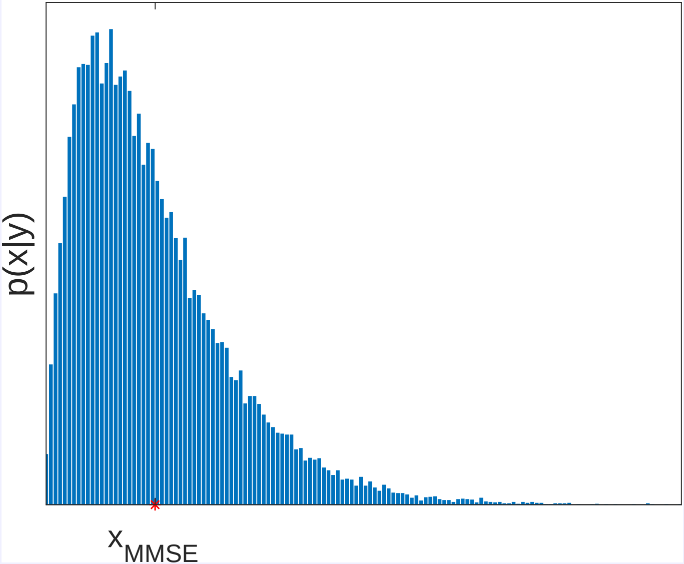
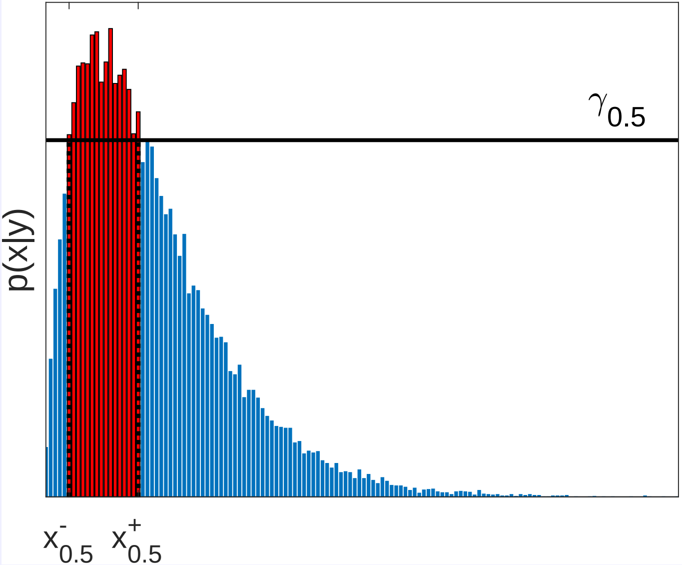


Monte Carlo methods for Bayesian inference
Approximating the MMSE estimator



Monte Carlo methods for Bayesian inference

Deriving credibility intervals (*highest posterior density*)



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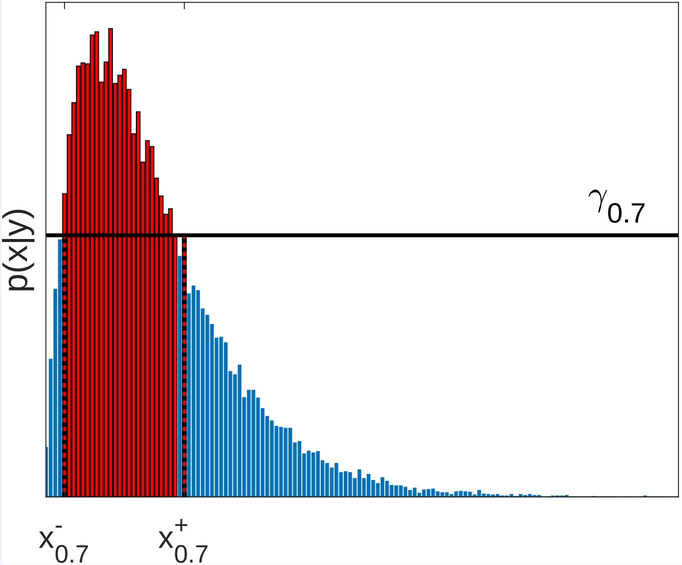

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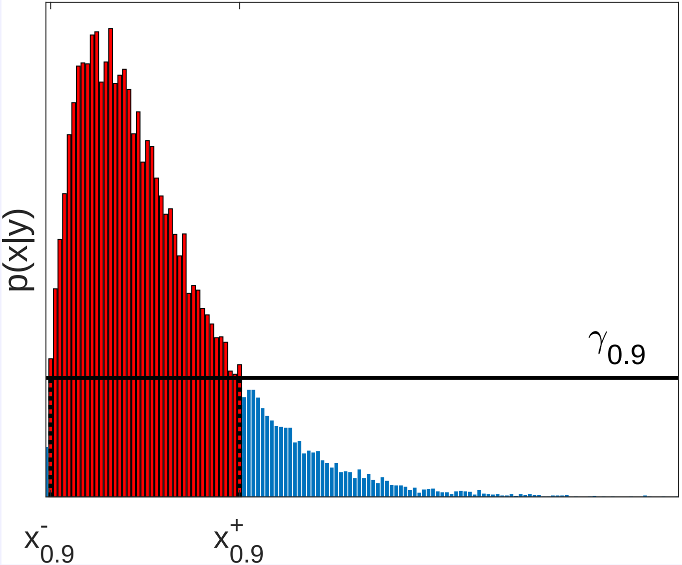

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Deriving credibility intervals (*highest posterior density*)

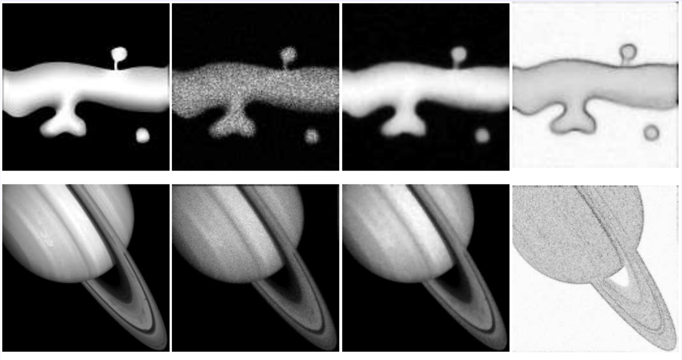

Figure: from Vono et al. (2018)

Generation of random variables

Direct sampling

For simple univariate and multivariate distributions:

- generation after inversion of the cumulative distribution function
  (e.g., exponential distribution)
- generation by change of variables
  (e.g., Box-Muller for univariate Gaussian distribution)
- ...

widely implemented in scientific computing softwares (Python, Matlab, R,...).

Non-standard sampling

Required once targeting

- non-standard univariate distributions
  (e.g., truncated distribution)
- non-standard multivariate distributions
  (e.g., posterior distribution resulting from a hierarchical Bayesian model)

Challenge

Drawing $\theta^{(1)}, \ldots, \theta^{(T)} \sim f(\theta)$.

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Outline

Importance sampling

Assumption

One cannot sample θ from $f(\cdot)$ but one knows how to sample from $q(\cdot)$ such that $\mathrm{supp}(q) \supset \mathrm{supp}(f)$.

Reformulation of the expectation

$$\begin{aligned}
\mathrm{E}_{\theta \sim f(\theta)}\left(G(\theta)\right) &= \int G(\theta) f(\theta) d\theta \\
&= \int G(\theta) \frac{f(\theta)}{q(\theta)} q(\theta) d\theta \\
&= \mathrm{E}_{\theta \sim q(\theta)}\left(G(\theta) \frac{f(\theta)}{q(\theta)}\right)
\end{aligned}$$

thus one can approximate $\mathrm{E}_f\left(G(\theta)\right)$ as

$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} G(\theta^{(t)}) \frac{f(\theta^{(t)})}{q(\theta^{(t)})}$$

where $\theta^{(t)}$ drawn from $q(\theta)$, referred to as **instrumental**/**proposal** distribution.

Accept-reject algorithm

Assumption

One cannot sample θ from $f(\cdot)$ but one knows how to sample from a *proposal* distribution $q(\cdot)$ for which it exists $M > 0$ such that, $\forall\theta$, $f(\theta) \leq Mq(\theta)$.

Algorithm

while $t < T$ **do**
  draw $\theta^* \sim q(\theta)$
  compute $\rho = \frac{f(\theta^*)}{Mq(\theta^*)}$
  draw $u \sim \mathcal{U}(0, 1)$
  if $u < \rho$ **then**
    $\theta^{(t)} \leftarrow \theta^*$ (accept)
    $t \leftarrow t + 1$
  else
    reject
  end if
end while

Properties

- The generated samples $\theta^{(1)}, \ldots, \theta^{(T)}$ are distributed according to $f(\theta)$
- The generated samples $\theta^{(1)}, \ldots, \theta^{(T)}$ are independent.
- The probability of acceptance is $\frac{1}{M}$.

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Metropolis-Hastings algorithm

Assumption

One cannot sample θ from $f(\cdot)$ but one knows how to sample from the proposal $q(\cdot|\cdot)$ such that $\mathrm{supp}(q) \supset \mathrm{supp}(f)$.

Iterative algorithm

Initialization: $\theta^{(0)}$
for $t = 1$ to T **do**
  draw $\theta^\star \sim q(\theta^\star|\theta^{(t-1)})$
  compute $\rho = \min\left\{1, \frac{f(\theta^\star)}{f(\theta^{(t-1)})}\frac{q(\theta^{(t-1)}|\theta^\star)}{q(\theta^\star|\theta^{(t-1)})}\right\}$
  draw $u \sim \mathcal{U}(0,1)$
  if $u < \rho$ **then**
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Metropolis-Hastings algorithm

Properties

- **seminal** instance of Markov chain Monte Carlo (MCMC) sampling.
- after a **burn-in period**, the generated samples $\theta^{(T_0)}, \ldots, \theta^{(T)}$ are asymptotically distributed according to $f(\cdot)$.
- the generated samples $\theta^{(1)}, \ldots, \theta^{(T)}$ are **not independent** (they build a Markov chain and, in particular, one may have $\theta^{(t)} = \theta^{(t-1)}$)

Application to Bayesian inference

The Metropolis-Hastings ratio writes

$$\rho = \min\left\{1, \frac{f(\mathbf{x}^\star|\mathbf{y})}{f(\mathbf{x}^{(t-1)}|\mathbf{y})} \frac{q(\mathbf{x}^{(t-1)}|\mathbf{x}^\star)}{q(\mathbf{x}^\star|\mathbf{x}^{(t-1)})}\right\}$$

thus the posterior distribution $f(\cdot|\mathbf{y})$ can be defined up to a normalizing constant.

$\rightarrow$ Good news, in particular within a Bayesian framework

$$f(\mathbf{x}|\mathbf{y}) = \frac{f(\mathbf{y}|\mathbf{x})f(\mathbf{x})}{f(\mathbf{y})}$$

where the evidence $f(\mathbf{y})$ is hard to compute (or even not known).

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Particular instances

Independent case

- proposal defined as $q(\cdot|\theta^{(t-1)}) = q(\cdot)$
- does not take the current state $\theta^{(t-1)}$ into account

Symmetric case

- proposal defined as $q(\theta^\star|\theta^{(t-1)}) = q(\theta^{(t-1)}|\theta^\star)$
- Metropolis-Hastings ratio defined as

$$\rho = \min\left\{1, \frac{f(\theta^\star)}{f(\theta^{(t-1)})}\right\}$$

  (e.g., Gaussian random walk)
- algorithmic behavior
  - if $f(\theta^\star) \geq f(\theta^{(t-1)})$, then $\theta^{(t)} \leftarrow \theta^\star$
  - if $f(\theta^\star) < f(\theta^{(t-1)})$, then $\theta^{(t)} \leftarrow \theta^\star$ with proba $\dfrac{f(\theta^\star)}{f(\theta^{(t-1)})}$

In the context of **Bayesian inference**

- if $\mathbf{x}^\star$ increases the posterior distribution $f(\mathbf{x}|\mathbf{y})$, then one accepts $\mathbf{x}^\star$
- if $\mathbf{x}^\star$ decreases the posterior distribution $f(\mathbf{x}|\mathbf{y})$, then one may still accept $\mathbf{x}^\star$

contrary to an optimization method looking for the MAP estimator!

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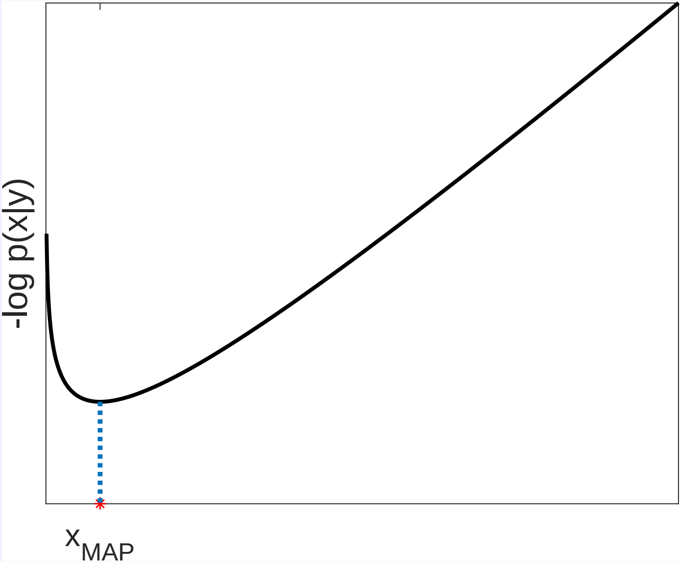

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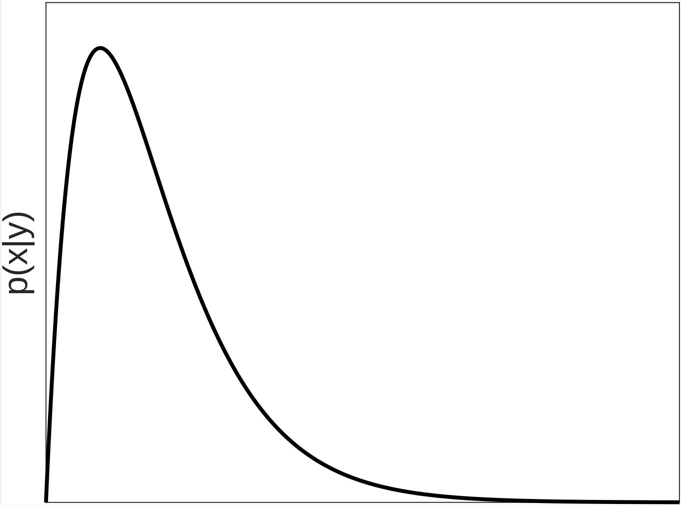

Gibbs sampling

Assumption
One want to draw $\boldsymbol{\theta}$ from $f(\boldsymbol{\theta})$ where

- the M-dimensional parameter $\boldsymbol{\theta}$ writes $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_M]^{\mathsf{T}}$
- the M conditional distributions $f\left(\theta_m | \boldsymbol{\theta}_{\setminus m}\right)$ are known with $\boldsymbol{\theta}_{\setminus m} = [\theta_1, \ldots, \theta_{m-1}, \theta_{m+1}, \ldots, \theta_M]^{\mathsf{T}}$

Principle
Iteratively sampling from the M **conditional distributions**.

Iterative algorithm

Initialization: $\boldsymbol{\theta}_{\setminus 1}^{(0)} = \left[\theta_2^{(0)}, \theta_3^{(0)}, \ldots, \theta_M^{(0)}\right]^{\mathsf{T}}$
for $t = 1$ to T **do**
  draw $\theta_1^{(t)} \sim f\left(\theta_1 | \theta_2^{(t-1)}, \theta_3^{(t-1)}, \ldots, \theta_M^{(t-1)}\right)$
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Gibbs sampling

Properties

- a special case of Metropolis-Hastings where the proposals are the conditional distributions
- the set of T generated samples $\left\{\boldsymbol{\theta}^{(t)}\right\}_{t=1}^{T} = \left\{\left(\theta_1^{(t)}, \ldots, \theta_M^{(t)}\right)\right\}_{t=1}^{T}$ are asymptotically distributed according to the joint distribution $f(\boldsymbol{\theta}) = f\left(\theta_1, \ldots, \theta_N\right)$
- the M sets of T generated samples $\left\{\theta_m^{(t)}\right\}_{t=1}^{T}$ are asymptotically distributed according to the M marginal distributions $f\left(\theta_m\right)$

Particular instances

- ***Block scanning**:* the conditional distributions can be defined from block of variables (not necessarily mono-dimensional)
- ***Random scanning**:* the order of the M updates can be randomly drawn at each iteration t
- ***Metropolis-within-Gibbs**:* one or several of these M sampling steps can be conducted thanks to a Metropolis-Hastings steps.

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Block coordinate descent / alternate optimization

Initialization: $\mathbf{x}_{\backslash 1}^{(0)} = \left[x_2^{(0)}, x_3^{(0)}, \ldots, x_M^{(0)}\right]^{\mathrm{T}}$

for $t = 1$ to T **do**

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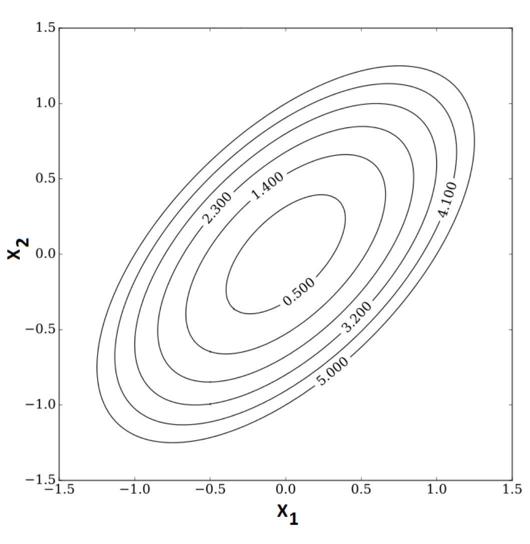

Coordinate descent.

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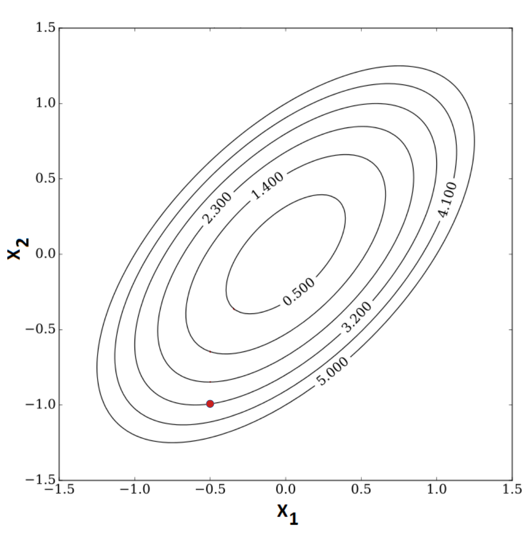

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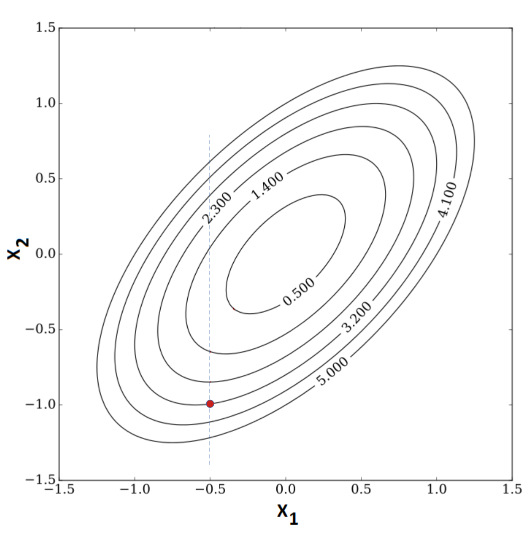

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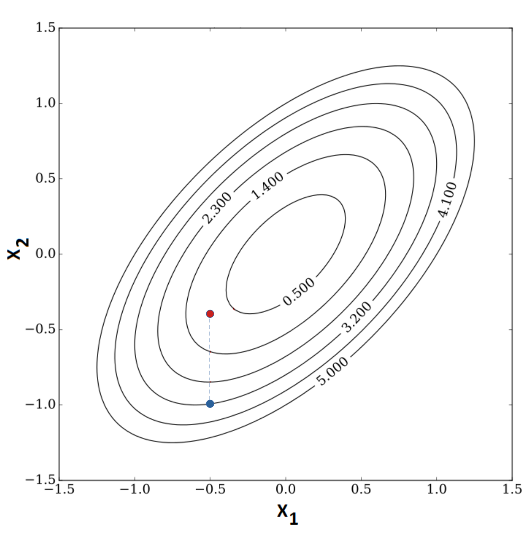

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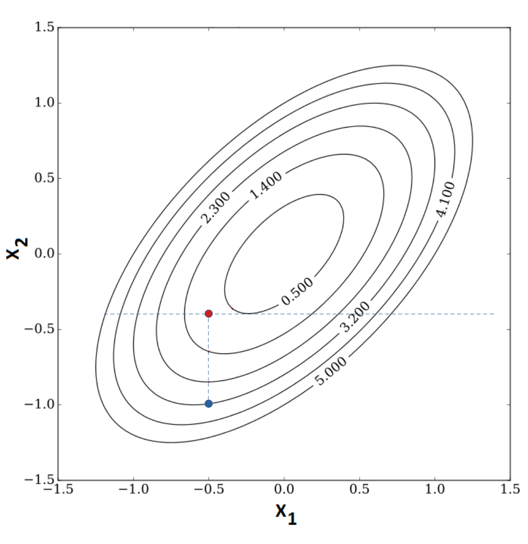

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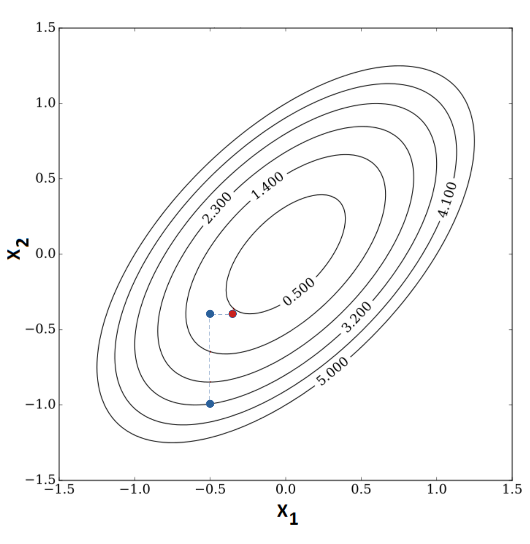

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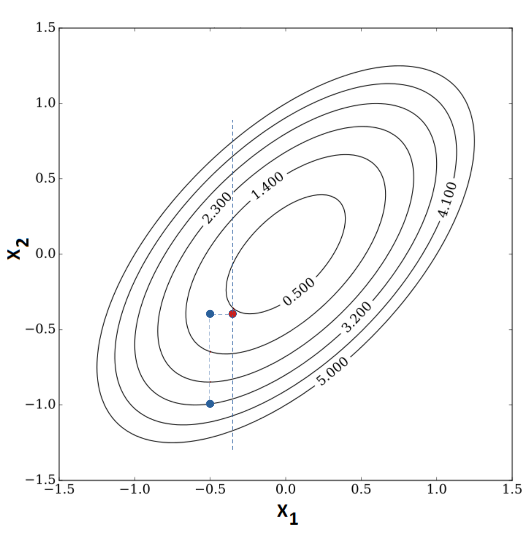

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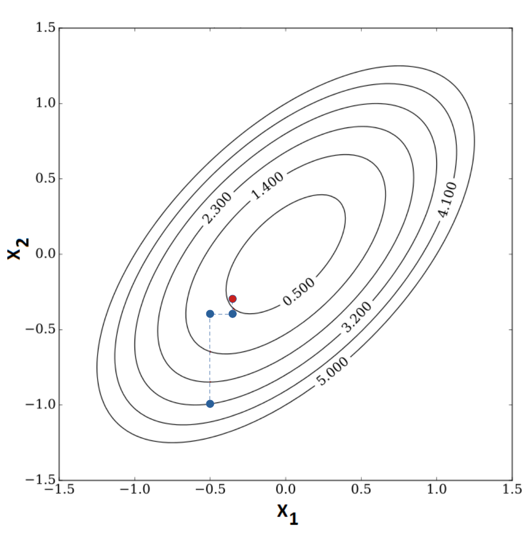

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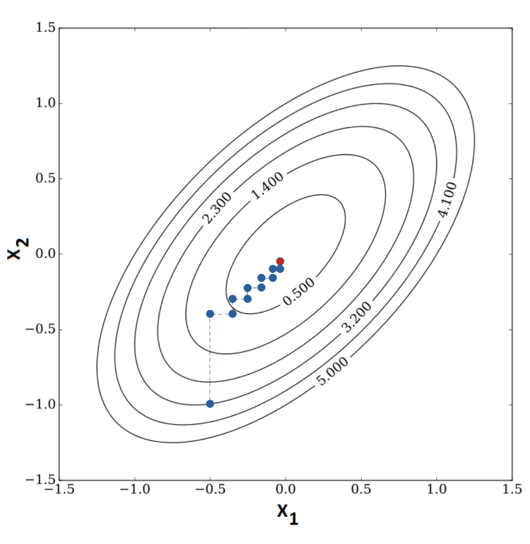

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Block coordinate descent / alternate optimization

Initialization: $\mathbf{x}_{\setminus 1}^{(0)} = \left[x_2^{(0)}, x_3^{(0)}, \ldots, x_M^{(0)}\right]^T$
for $t = 1$ to T **do**
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Gibbs sampling
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Block coordinate descent / alternate optimization (MAP estimation)

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for $t = 1$ to T **do**
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$\rightarrow$ Output: $\mathbf{x}^{(T)}$ approximates $\hat{\mathbf{x}}_{\mathrm{MAP}}$.

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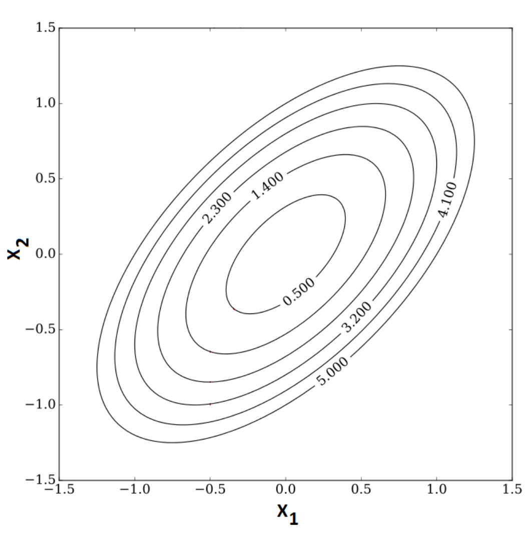
Gibbs sampling
Sampling vs. optimizing



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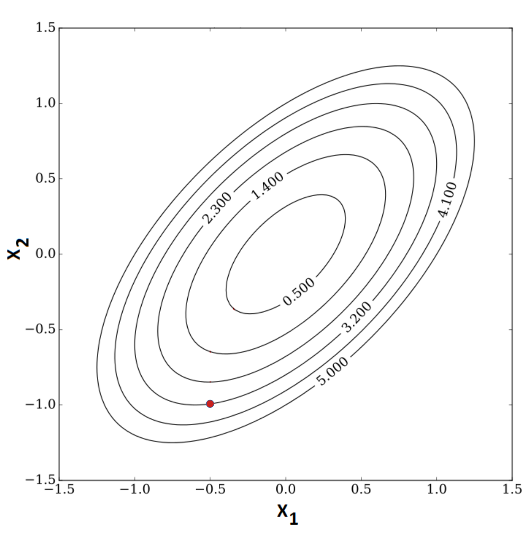
Gibbs sampling
Sampling vs. optimizing



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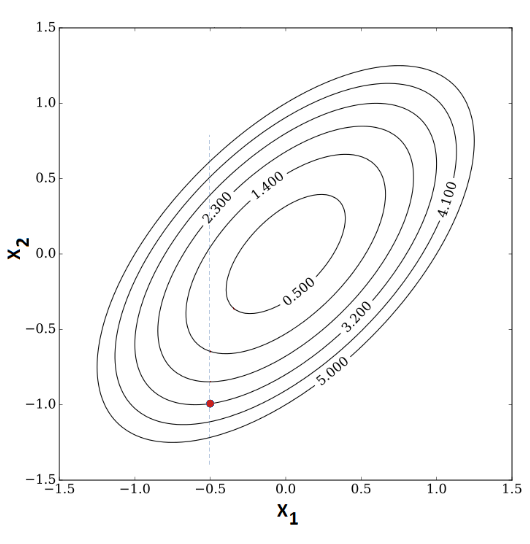

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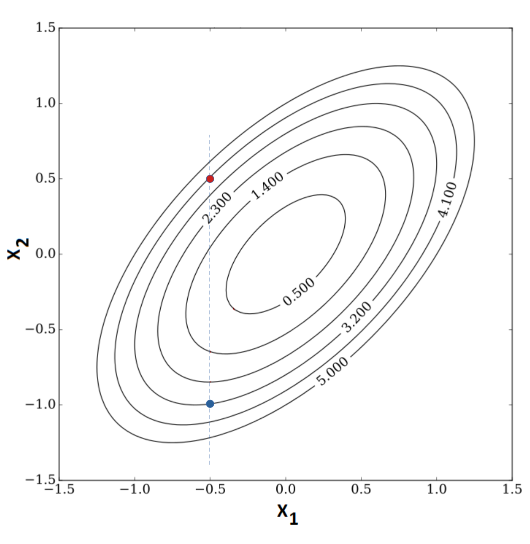

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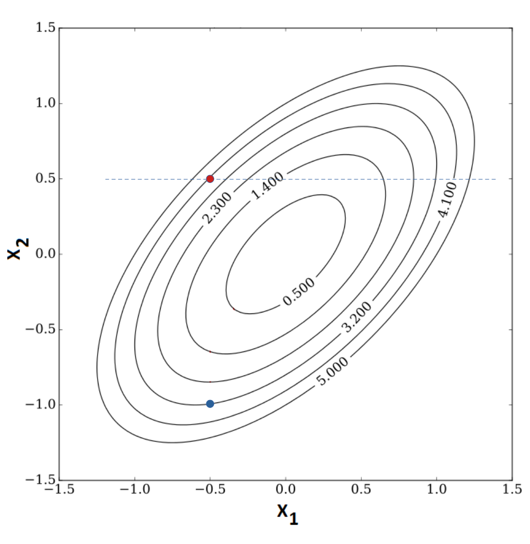

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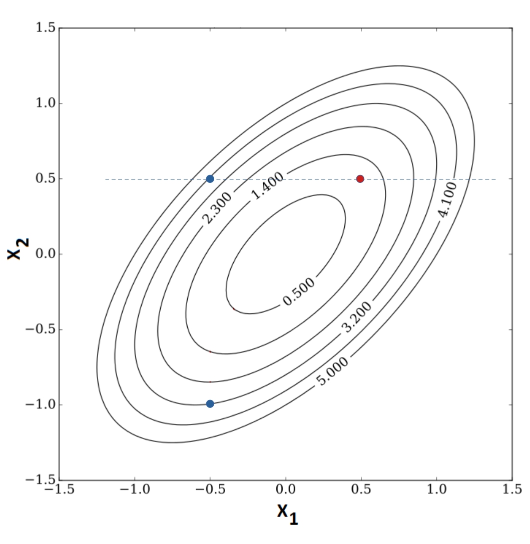

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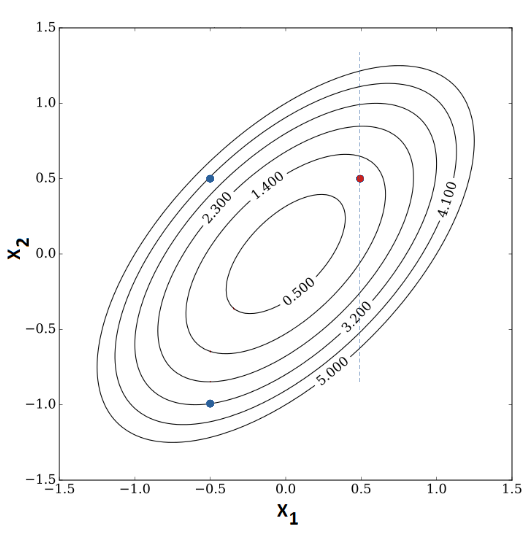

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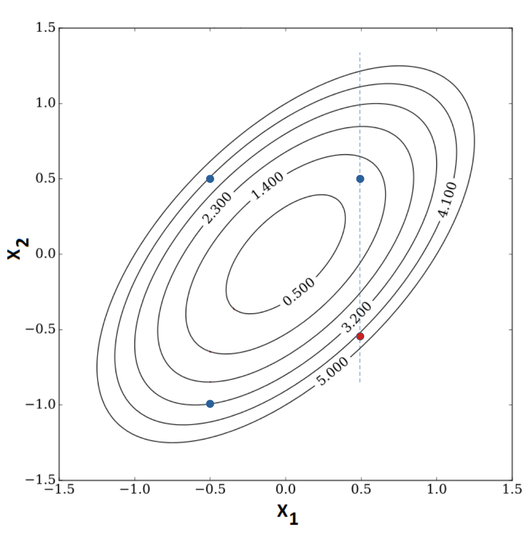

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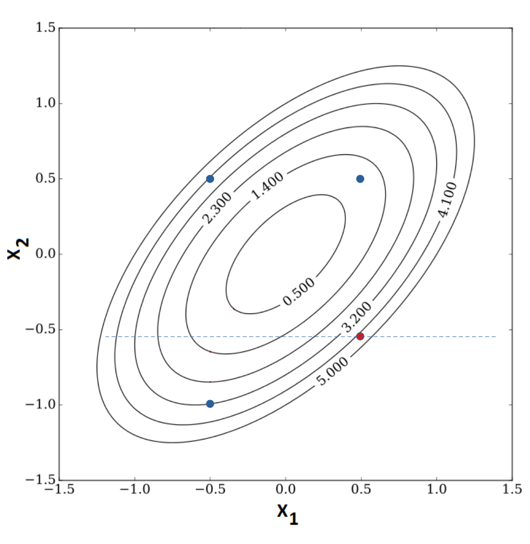

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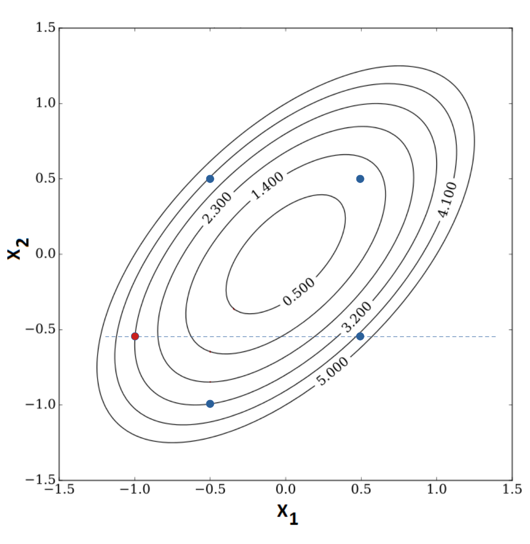

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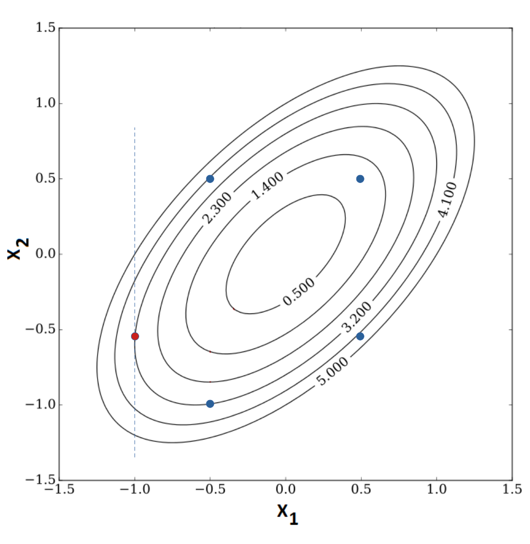

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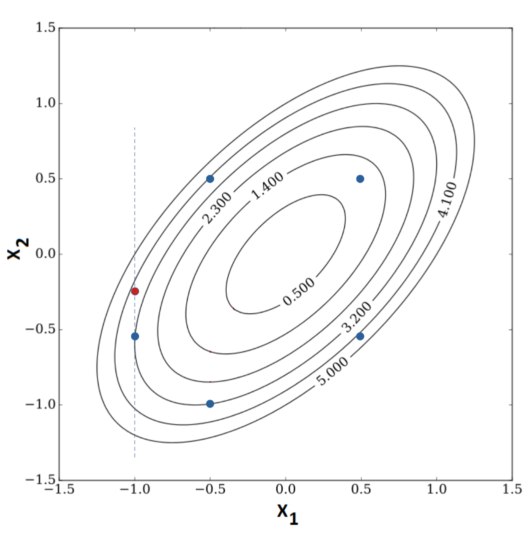

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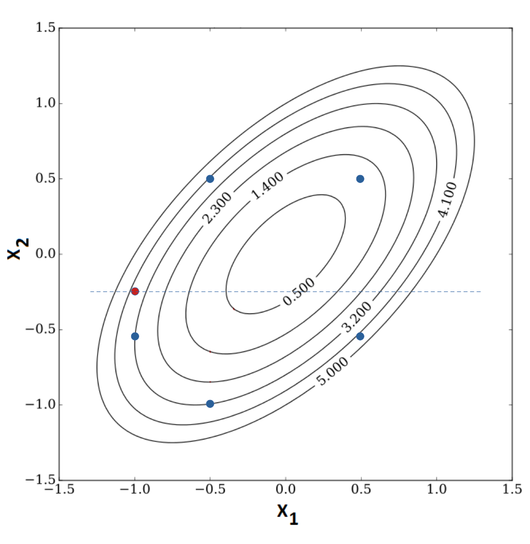

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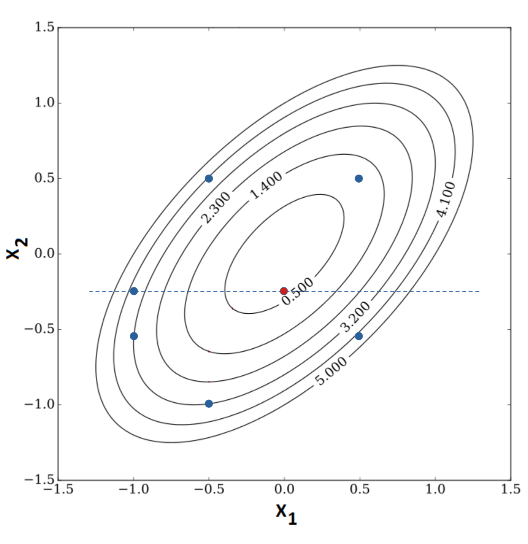

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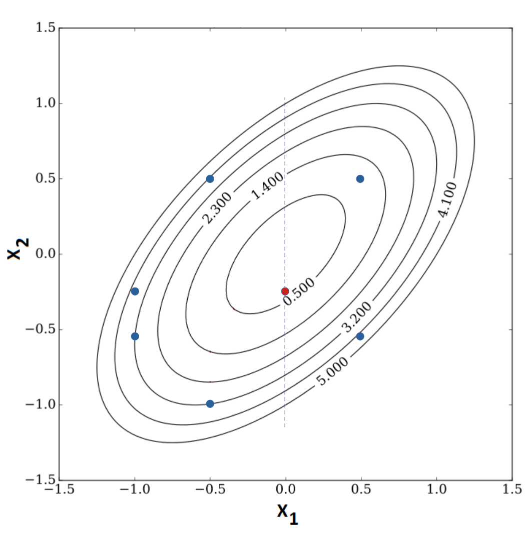

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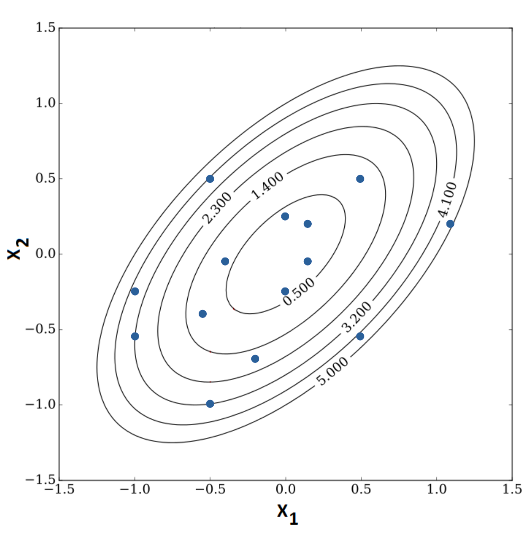

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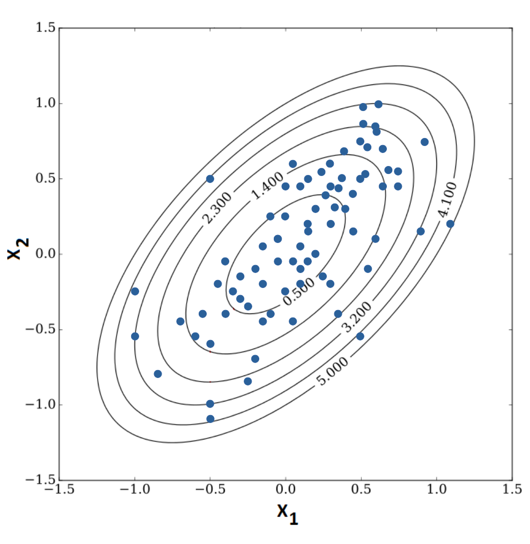

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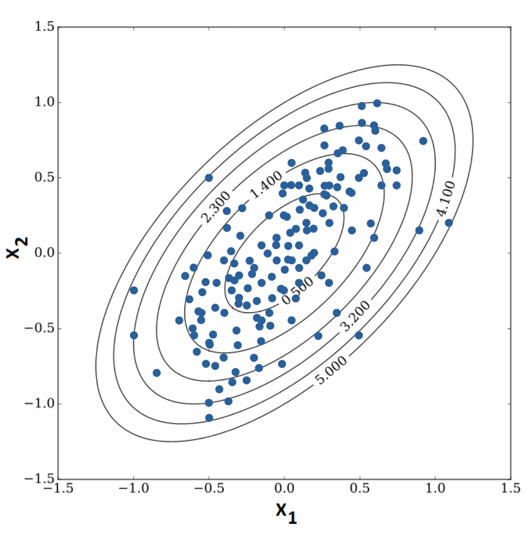

Gibbs sampling.

Gibbs sampling
Sampling vs. optimizing

Application to Bayesian inference

- the unknown parameter $\mathbf{x}$ writes $\mathbf{x} = [x_1, \ldots, x_M]^{\mathrm{T}}$
- the posterior distribution $p(\mathbf{x}|\mathbf{y})$ writes $p(\mathbf{x}|\mathbf{y}) = p(x_1, \ldots, x_M|\mathbf{y})$

Gibbs sampling

Initialization: $\mathbf{x}_{\backslash 1}^{(0)} = \left[x_2^{(0)}, x_3^{(0)}, \ldots, x_M^{(0)}\right]^{\mathrm{T}}$
for $t = 1$ to T **do**
  draw $x_1^{(t)} \sim f\left(x_1|x_2^{(t-1)}, x_3^{(t-1)}, \ldots, x_M^{(t-1)}, \mathbf{y}\right)$
  draw $x_2^{(t)} \sim f\left(x_2|x_1^{(t)}, x_3^{(t-1)}, \ldots, x_M^{(t-1)}, \mathbf{y}\right)$
  ...
  draw $x_M^{(t)} \sim f\left(x_M|x_1^{(t)}, \ldots, x_{M-1}^{(t)}, \mathbf{y}\right)$
end for

Gibbs sampling

Sampling vs. optimizing

Application to Bayesian inference

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- the posterior distribution $p(\mathbf{x}|\mathbf{y})$ writes $p(\mathbf{x}|\mathbf{y}) = p(x_1, \ldots, x_M|\mathbf{y})$

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...

draw $x_M^{(t)} \sim f\left(x_M | x_1^{(t)}, \ldots, x_{M-1}^{(t)}, \mathbf{y}\right)$

end for

$\rightarrow$ Output: $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(T)}$ distributed according to $p(\mathbf{x}|\mathbf{y})$.

Gibbs sampling
Application to (exact) data augmentation

Objective

One aims at sampling from $f(\theta|\phi)$

- which is not simple
- but writes $f(\theta|\phi) = \int f(\theta, z|\phi)\, dz$

where the distributions

- $f(z|\theta, \phi)$
- $f(\theta|z, \phi)$

are simple to sample from.

Idea: sampling from the augmented distribution $f(\theta, z|\phi)$

Gibbs sampling writes

- $\theta \sim f(\theta|z, \phi)$
- $z \sim f(z|\theta, \phi)$

Properties

- the generated samples $\theta^{(1)}, \ldots, \theta^{(T)}$ are distributed according to $f(\theta)$
- when $f(\theta, z) = f(\theta|z)\, f(z)$, one performs **ancestral sampling**

Ancestral sampling

Application to discrete mixture distributions

One aims at sampling from $f(\theta)$ defined as a **mixture of distributions**

$$f(\theta) = (1-\pi) f_0(\theta) + \pi f_1(\theta)$$

where $f_0(\theta)$ and $f_1(\theta)$ are easy to sample from.

Augmented distribution

- One introduces the binary variable z such that

$$\mathrm{P}[z=0] = 1-\pi \qquad \text{and} \qquad \mathrm{P}[z=1] = \pi$$

- The augmented distribution writes

$$f(\theta, z) = f(\theta|z)\, f(z)$$

which ensures

$$f(\theta) = \sum_{k\in\{0,1\}} f(\theta|z=k)\, \mathrm{P}[z=k])$$

with $f(\theta|z=k) = f_k(\theta)$.

Ancestral sampling

- $z \sim \mathrm{Bernoulli}(\pi)$
- $\theta \sim \begin{cases} f_0(\theta) & \text{if} \quad z=0, \\ f_1(\theta) & \text{if} \quad z=1. \end{cases}$

Gibbs sampling
Application to Bayesian inference

Bayesian model

- likelihood function $f(\mathbf{y}|\mathbf{x}, \mathbf{z})$

and assuming prior independence between $\mathbf{z}$ and $\mathbf{x}$

- parameter prior distribution $f(\mathbf{x})$
- (nuisance) parameter prior distribution $f(\mathbf{z})$

Joint posterior distribution

Defined as

$$f(\mathbf{x}, \mathbf{z}|\mathbf{y}) \propto f(\mathbf{y}|\mathbf{x}, \mathbf{z}) f(\mathbf{x}) f(\mathbf{z})$$

Gibbs sampling

$$\begin{aligned} \mathbf{x}^{(t)} &\sim f\left(\mathbf{x}|\mathbf{z}^{(t-1)}, \mathbf{y}\right) \\ \mathbf{z}^{(t)} &\sim f\left(\mathbf{z}|\mathbf{x}^{(t)}, \mathbf{y}\right) \end{aligned}$$

MMSE estimators

$$\hat{\mathbf{x}}_{\mathrm{MMSE}} \approx \frac{1}{T}\sum_{t=1}^{T} \mathbf{x}^{(t)} \quad \text{and} \quad \hat{\mathbf{z}}_{\mathrm{MMSE}} \approx \frac{1}{T}\sum_{t=1}^{T} \mathbf{z}^{(t)}$$

Gibbs sampling

Application to Bayesian inference (example)

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{b} \quad \text{with} \quad \mathbf{b} \sim \mathcal{N}\left(\mathbf{0}, \lambda \mathbf{I}\right)$$

Bayesian model with conjugate priors

- likelihood function $\mathbf{y}|\mathbf{x}, \sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$
- parameter prior distribution $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}}, \sigma^2_{\mathbf{x}} \mathbf{I})$
- (nuisance) parameter prior distribution $\sigma^2 \sim \mathcal{IG}(\alpha, \beta)$

Joint posterior distribution

Defined as

$$\begin{aligned} f(\mathbf{x}, \sigma^2|\mathbf{y}) &\propto f(\mathbf{y}|\mathbf{x}, \sigma^2) f(\mathbf{x}) f(\sigma^2) \\ &\propto \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp\left(-\frac{1}{\sigma^2}\left\|\mathbf{y} - \mathbf{H}\mathbf{x}\right\|_2^2 - \frac{1}{2\sigma^2_{\mathbf{x}}}\left\|\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}}\right\|_2^2\right) \left(\frac{1}{\sigma^2}\right)^{\alpha} \exp\left(-\frac{\beta}{\sigma^2}\right) \end{aligned}$$

Gibbs sampling

$$\begin{aligned} \mathbf{x}^{(t)} &\sim \mathcal{N}\left(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}}, \boldsymbol{\Sigma}^{(t-1)}_{\mathbf{x}|\mathbf{y}}\right) \\ \sigma^{2(t)} &\sim \mathcal{IG}\left(\alpha + \frac{1}{2}\|\mathbf{y} - \mathbf{H}\mathbf{x}^{(t)}\|_2^2, \beta + \frac{n}{2}\right) \end{aligned}$$

Gibbs sampling
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Gibbs sampling

$$\begin{aligned} \mathbf{x}^{(t)} &\sim \mathcal{N}\left(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}}, \boldsymbol{\Sigma}_{\mathbf{x}|\mathbf{y}}^{(t-1)}\right) \\ \sigma^{2(t)} &\sim \mathcal{IG}\left(\alpha + \frac{1}{2}\|\mathbf{y} - \mathbf{H}\mathbf{x}^{(t)}\|_2^2, \beta + \frac{n}{2}\right) \end{aligned}$$

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Gibbs sampling
Application to hierarchical Bayesian inference

Hierarchical Bayesian model

- likelihood function $f(\mathbf{y}|\mathbf{x})$
- parameter prior distribution $f(\mathbf{x}|\phi)$
- hyperparameter prior distribution $f(\phi)$

Joint posterior distribution
Defined as

$$f(\mathbf{x}, \phi|\mathbf{y}) \propto f(\mathbf{y}|\mathbf{x})f(\mathbf{x}|\phi)f(\phi)$$

Gibbs sampling

$$\begin{aligned} \mathbf{x}^{(t)} &\sim f\left(\mathbf{x}|\phi^{(t-1)}, \mathbf{y}\right) \\ \phi^{(t)} &\sim f\left(\phi|\mathbf{x}^{(t)}\right) \end{aligned}$$

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Hierarchical Bayesian model with conjugate priors

- likelihood function $\mathbf{y}|\mathbf{x}, \sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$
- parameter prior distribution $\mathbf{x}|\boldsymbol{\mu}_{\mathbf{x}} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}}, \sigma_{\mathbf{x}}^2\mathbf{I})$
- hyperparameter prior distribution $\boldsymbol{\mu}_{\mathbf{x}} \sim \mathcal{N}(\boldsymbol{\mu}_0, \sigma_0^2\mathbf{I})$

Joint posterior distribution
Defined as

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Gibbs sampling

$$\begin{aligned} \mathbf{x}^{(t)} &\sim \mathcal{N}\left(\mu_{\mathbf{x}|\mathbf{y}}^{(t-1)}, \Sigma_{\mathbf{x}|\mathbf{y}}^{(t-1)}\right) \\ \mu_{\mathbf{x}}^{(t)} &\sim \mathcal{N}\left(\mu_{\mathbf{x}|\mathbf{x}}^{(t)}, \sigma_{\mathbf{x}|\mathbf{x}}^2\mathbf{I}\right) \end{aligned}$$

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Outline

Current limitations

Metropolis-Hastings (MH) algorithm

The proposal distribution drives the mixing properties, a trade-off between

- minimizing the sample correlations and correctly explore the solution space
- ensuring an “admissible” acceptance ratio

Gibbs sampling

Component-by-component updating rules have some advantages

- decrease the dimension of the space to be explored (e.g., updating one pixel in an image)
- ease the choice of the proposal distribution in case of Metropolis-within-Gibbs

but comes with serious shortcomings

- known/computable conditional distributions
- poor mixing properties (high correlation between generated samples)
- high computational cost (*for loop* within a *for loop*...)

Opportunities

- building on concepts from **computational/statistical physics**
- translating algorithmic tricks and methodological tools from **optimization**
- benefiting from recent advances in **deep (generative) modeling**

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Exact Perturbation Optimization (E-PO)

Framework: Bayesian linear models

- Likelihood function: $\mathbf{y} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ with $\mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$
- Prior distribution: $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}})$

One aims at sampling from the posterior Gaussian distribution

$$f(\boldsymbol{\theta}|\mathbf{y}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}})^{\mathsf{T}}\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}})\right]$$

with

$$\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}} = \mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}\mathbf{H} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}} = \mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}^{-1}\left[\mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{w}}) + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}\boldsymbol{\mu}_{\boldsymbol{\theta}}\right]$$

Major issues

- Inversion of $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}$
- The vanilla Gaussian sampler requires the Cholesky decomposition of $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}^{-1}$

E-PO algorithm

- Draw $\mathbf{z}_{\mathbf{w}} \sim \mathcal{N}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$
- Draw $\mathbf{z}_{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}})$
- Set $\eta = \mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}\mathbf{z}_{\mathbf{w}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}\mathbf{z}_{\boldsymbol{\theta}}$
- Solve the linear problem $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}\theta = \eta$

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- Prior distribution: $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}})$

One aims at sampling from the posterior Gaussian distribution

$$f(\boldsymbol{\theta}|\mathbf{y}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}})^{\mathsf{T}}\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}(\boldsymbol{\theta} - \boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}})\right]$$

with

$$\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}} = \mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}\mathbf{H} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta}|\mathbf{y}} = \mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}^{-1}\left[\mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{w}}) + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}\boldsymbol{\mu}_{\boldsymbol{\theta}}\right]$$

Major issues

- Inversion of $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}$
- The vanilla Gaussian sampler requires the Cholesky decomposition of $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}^{-1}$

E-PO algorithm

- Draw $\mathbf{z}_{\mathbf{w}} \sim \mathcal{N}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$
- Draw $\mathbf{z}_{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}})$
- Set $\boldsymbol{\eta} = \mathbf{H}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}\mathbf{z}_{\mathbf{w}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}\mathbf{z}_{\boldsymbol{\theta}}$
- Solve the linear problem $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}\boldsymbol{\theta} = \boldsymbol{\eta}$

Truncated & Reversible Jump Perturbation Optimization (T-PO & RJ-PO)

Remaining issue
Solving the linear problem

$$\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}}\boldsymbol{\theta} = \boldsymbol{\eta}$$

with $\mathbf{Q}_{\boldsymbol{\theta}|\mathbf{y}} = \mathbf{H}^T\boldsymbol{\Sigma}_{\mathbf{w}}^{-1}\mathbf{H} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}}$.

Truncated Perturbation Optimization (T-PO)

- Using an iterative *solver*, e.g., (early stopped) conjugate gradient
- Provides a sample θ^* drawn from an approximation of the target distribution

Reversible Jump Perturbation Optimization (RJ-PO)

- Approximation mitigated by a Metropolis-Hastings step with acceptance ratio

$$\rho = \min\left\{1, \exp\left[\left(\mathbf{Q}_{\theta|\mathbf{y}}\theta^* - \mu\right)(\theta - \theta^*)\right]\right\}$$

Remark: wide literature dedicated to particular cases, e.g., $\mathbf{Q}_{\theta|\mathbf{y}} = \mathbf{Q}_1 + \mathbf{Q}_2$ *with structured matrices* $\mathbf{Q}_1$ *and* $\mathbf{Q}_2$ *(e.g., diagonalizable in in the same basis).*

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Hamiltonian Monte Carlo

Objective

One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$.

Idea: extended model

- Let consider the Hamiltonian associated with $U(\cdot)$
$$H(\boldsymbol{\theta}, \mathbf{p}) = U(\boldsymbol{\theta}) + K(\mathbf{p})$$
where $K(\mathbf{p})$ is a kinetic energy function, e.g., $K(\mathbf{p}) = \frac{\mathbf{p}^T\mathbf{p}}{2}$
- Simulation of Hamiltonian dynamics after discretizing the stochastic differential equation
- Sampling from the joint distribution
$$\begin{aligned} h(\boldsymbol{\theta}, \mathbf{p}) &\propto \exp\left[-H(\boldsymbol{\theta}, \mathbf{p})\right] \\ &\propto \exp\left[-U(\boldsymbol{\theta}) - \frac{1}{2}\|\mathbf{p}\|^2\right] \end{aligned}$$
thanks to a Metropolis-Hastings algorithm.

Hamiltonian Monte Carlo

Drawing the proposed candidate $(\boldsymbol{\theta}^*, \mathbf{p}^*)$

- Initialization: $\mathbf{p}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:
- *Leapfrogs* thanks to N_{LF} iterations of the algorithmic scheme

$$\mathbf{p}^{(n+\frac{1}{2})} = \mathbf{p}^{(n)} - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}^{(n)}) \tag{1}$$

$$\boldsymbol{\theta}^{(n+\frac{1}{2})} = \boldsymbol{\theta}^{(n)} + \epsilon\mathbf{p}^{(n+\frac{1}{2})} \tag{2}$$

$$\mathbf{p}^{(n+1)} = \boldsymbol{\theta}^{(n+\frac{1}{2})} - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}^{(n+1)}) \tag{3}$$

- One proposes the candidate $(\boldsymbol{\theta}^*, \mathbf{p}^*) = (\boldsymbol{\theta}^{(N_{\mathrm{LF}})}, \mathbf{p}^{(N_{\mathrm{LF}})})$

Properties

- One is sampling according to $h(\boldsymbol{\theta}, \mathbf{p}) = f(\boldsymbol{\theta})\mathcal{N}(\mathbf{p}; \mathbf{0}, \mathbf{I})$
- In particular, the generated samples $\left\{\boldsymbol{\theta}^{(t)}\right\}_{t=1}^{T}$ are distributed according to $f(\cdot)$.

Difficulties

- How to choose ϵ and N_{LF}...

Langevin diffusion
Metropolis adjusted Langevin algorithm (MALA)

Objective
One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$.

Metropolis Adjusted Langevin algorithm (MALA)
When $N_{\mathrm{LF}} = 1$ in the HMC scheme, the candidates writes

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}^{(t)} - \delta \nabla U(\boldsymbol{\theta}^{(t)}) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

with $\delta = \frac{\epsilon^2}{2}$ and $\mathbf{p}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

- Euler-Maruyama time-discrete approximation of a Langevin diffusion
- the approximation is mitigated by a Metropolis-Hastings (MH) step

Langevin diffusion
Unajusted Langevin algorithms (ULA)

Objective
One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$.

Unajusted Langevin algorithm (ULA)
When deciding to bypass the MH step in MALA, the updating rule writes

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \delta \nabla U(\boldsymbol{\theta}^{(t)}) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

with $\delta = \frac{\epsilon^2}{2}$ and $\mathbf{p}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

- avoiding the computation burden required by the MH step
- targeting an approximation of $f(\cdot)$ (with some control)

Remarks (HMC, MALA, ULA)

- Sampling driven by the gradient (towards the high probability density regions)
- Generalization of HMC and MALA with pre-conditionning, e.g.,

$$\boldsymbol{\theta}^{*} = \boldsymbol{\theta}^{(t)} - \delta \mathbf{G}^{-1}(\boldsymbol{\theta}^{(t)}) \nabla U(\boldsymbol{\theta}^{(t)}) + \sqrt{2\delta \mathbf{G}^{-1}(\boldsymbol{\theta}^{(t)})}\mathbf{p}^{(t)}$$

  where $\mathbf{G}(\cdot)$ is a metric related to an appropriate manifold (FIM, Hessian, ...),
- Requires $U(\cdot)$ to be smooth...

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Proximal Metropolis adjusted Langevin algorithm (P-MALA)

Objective
One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$ for a convex but non-smooth $U(\cdot)$.

Idea: Moreau-Yoshida approximation of $U(\cdot)$
The target distribution $f(\cdot)$ is approximated by

$$f_\lambda(\boldsymbol{\theta}) \propto \exp\left[-U_\lambda(\boldsymbol{\theta})\right]$$

where $U_\lambda(\cdot)$ is the **Moreau-Yoshida envelope** of $U(\cdot)$

$$U_\lambda(\boldsymbol{\theta}) = \inf_{\mathbf{v}} \left\{ U(\mathbf{v}) + \frac{1}{2\lambda} \|\boldsymbol{\theta} - \mathbf{v}\|_2^2 \right\}$$

and λ adjusts the approximation (w.r.t. pointwise convergence)

$$\lim_{\lambda \to 0} f_\lambda(\boldsymbol{\theta}) = f(\boldsymbol{\theta}) \qquad (\forall \boldsymbol{\theta}).$$

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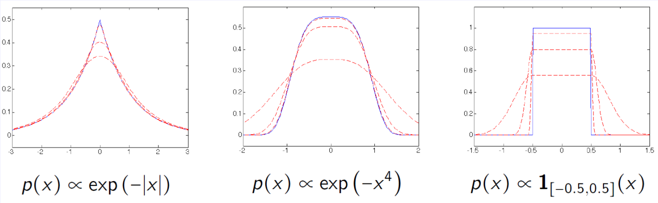

From [Pereyra2016].

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Properties

- $f_\lambda(\cdot)$ defines a probability density function
- $U_\lambda(\cdot)$ is gradient Lipchitz with

$$\nabla U_\lambda(\boldsymbol{\theta}) = \frac{1}{\lambda} \left\{ \boldsymbol{\theta} - \mathrm{prox}_U^\lambda(\boldsymbol{\theta}) \right\}$$

Proximal Metropolis adjusted Langevin algorithm (P-MALA)

Metropolis adjusted Langevin algorithm to target $f_\lambda(\cdot)$
Using

$$\nabla U_\lambda(\boldsymbol{\theta}) = \frac{1}{\lambda}\left(\boldsymbol{\theta} - \mathrm{prox}_U^\lambda(\boldsymbol{\theta})\right)$$

the proposal rule writes

$$\boldsymbol{\theta}^* = \left(1 - \frac{\delta}{\lambda}\right)\boldsymbol{\theta}^{(t)} - \frac{\delta}{\lambda}\mathrm{prox}_U^\lambda(\boldsymbol{\theta}^{(t)}) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

with $\mathbf{p}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Remarks

- Requires to mitigate the approximation by a Metropolis-Hastings step
  - increasing the computational cost
- Requires to know how to compute $\mathrm{prox}_U^\lambda(\cdot)$
  - not always straightforward, e.g., $U(\cdot) = E(\cdot) + R(\cdot)$

  but when $E(\cdot)$ is smooth, one approximation is

$$\boldsymbol{\theta}^* = \mathrm{prox}_R^\lambda\left(\boldsymbol{\theta}^{(t)} - \delta\nabla E(\boldsymbol{\theta}^{(t)})\right) + \sqrt{2\delta}\mathbf{p}$$

  (similar to a perturbed *forward-backard* scheme)

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Moreau-Yoshida regularised ULA (MYULA)

Objective

One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$ with $U(\cdot) = E(\cdot) + R(\cdot)$, where $E(\cdot)$ and $R(\cdot)$ are convex, $E(\cdot)$ is L_E-gradient Lipschitz but $R(\cdot)$ is not smooth.

Idea: Moreau-Yoshida approximation of $R(\cdot)$

The target distribution $f(\cdot)$ is approximated by

$$f_\lambda(\boldsymbol{\theta}) \propto \exp\left[-E(\boldsymbol{\theta}) - R_\lambda(\boldsymbol{\theta})\right]$$

where $R_\lambda(\cdot)$ is the **Moreau-Yoshida envelope** of $R(\cdot)$

$$R_\lambda(\boldsymbol{\theta}) = \inf_{\mathbf{v}} \left\{ R(\mathbf{v}) + \frac{1}{2\lambda} \|\boldsymbol{\theta} - \mathbf{v}\|_2^2 \right\}$$

and λ adjusts the approximation (w.r.t. TV convergence):

$$\lim_{\lambda \to 0} \|f - f_\lambda\|_{\mathrm{TV}} = 0.$$

Properties

- $f_\lambda(\cdot)$ defines a probability density function
- $U_\lambda(\cdot) = E(\cdot) + R_\lambda(\cdot)$ is gradient Lipchitz with

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Moreau-Yoshida regularised ULA (MYULA)

Unadjusted Langevin algorithm to target $f_\lambda(\cdot)$
Using

$$\nabla R_\lambda(\boldsymbol{\theta}) = \frac{1}{\lambda}\left(\boldsymbol{\theta} - \mathrm{prox}_R^\lambda(\boldsymbol{\theta})\right)$$

the updating rule writes

$$\boldsymbol{\theta}^{(t+1)} = \left(1 - \frac{\delta}{\lambda}\right)\boldsymbol{\theta}^{(t)} - \delta\nabla E(\boldsymbol{\theta}^{(t)}) + \frac{\delta}{\lambda}\mathrm{prox}_R^\lambda(\boldsymbol{\theta}^{(t)}) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

with $\mathbf{p}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Remarks

- Well suited for Bayesian inverse problem with non-smooth regularizations
- Mixing properties driven by the quantity $L_E + \lambda^{-1}$
  (as a trade-off between speed and approximation)
- When $E(\cdot)$ and $R(\cdot)$ are non-convex $\rightarrow$ *Forward-Backward Langevin Algorithm*

$$\boldsymbol{\theta}^{(t+1)} = \left(1 - \frac{\delta}{\lambda}\right)\boldsymbol{\theta}^{(t)} + \frac{\delta}{\lambda}\mathrm{prox}_R^\lambda\left(\boldsymbol{\theta}^{(t)} - \lambda\nabla E(\boldsymbol{\theta}^{(t)})\right) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

similar to a perturbed *forward-backward proximal splitting*.

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Plug-and-play ULA (PnP-ULA)

From Bayesian denoising...

$$\boldsymbol{\theta} = \mathbf{v} + \mathbf{b} \quad \text{with} \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$$

Bayesian model

- Likelihood function: $\boldsymbol{\theta}|\mathbf{v} \sim f(\boldsymbol{\theta}|\mathbf{v}) \propto \exp[-\frac{1}{2\lambda}\|\boldsymbol{\theta} - \mathbf{v}\|_2^2]$
- Prior distribution: $\mathbf{v} \sim f(\mathbf{v}) \propto \exp[-R(\mathbf{v})]$
- Posterior distribution defined as

$$f(\mathbf{v}|\boldsymbol{\theta}) = \frac{1}{f(\boldsymbol{\theta})} \exp\left[-\frac{1}{2\lambda}\|\boldsymbol{\theta} - \mathbf{u}\|_2^2 - R(\mathbf{v})\right]$$

with

$$f(\boldsymbol{\theta}) \propto \int \exp\left[-\frac{1}{2\lambda}\|\boldsymbol{\theta} - \mathbf{v}\|_2^2 - R(\mathbf{v})\right] d\mathbf{v}$$

Bayesian inference

- MMSE denoiser defined as $\mathrm{D}_\lambda^*(\boldsymbol{\theta}) = \hat{\mathbf{v}}_{\mathrm{MMSE}} = \mathrm{E}[\mathbf{v}|\boldsymbol{\theta}]$
- Tweedie formula

$$\mathrm{D}_\lambda^*(\boldsymbol{\theta}) = \boldsymbol{\theta} + \lambda \frac{\nabla f(\boldsymbol{\theta})}{f(\boldsymbol{\theta})}$$

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$$\mathrm{D}_\lambda^*(\boldsymbol{\theta}) = \boldsymbol{\theta} + \lambda \frac{\nabla f(\boldsymbol{\theta})}{f(\boldsymbol{\theta})} \quad \text{or} \quad \nabla R_\lambda(\boldsymbol{\theta}) = \frac{\mathrm{D}_\lambda^*(\boldsymbol{\theta}) - \boldsymbol{\theta}}{\lambda}$$

with

$$R_\lambda(\boldsymbol{\theta}) = \log \int \exp\left[-\frac{1}{2\lambda}\|\boldsymbol{\theta} - \mathbf{v}\|_2^2 - R(\mathbf{v})\right] d\mathbf{v} + \text{const.}$$

Plug-and-play ULA (PnP-ULA)

From Bayesian denoising...

Objective

One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$ with $U(\cdot) = E(\cdot) + R(\cdot)$ where $E(\cdot)$ is smooth and $R(\cdot)$ may be not specified.

Idea: smooth approximation of $R(\cdot)$

The target distribution $f(\cdot)$ is approximated by

$$f_\lambda(\boldsymbol{\theta}) \propto \exp\left[-E(\boldsymbol{\theta}) - R_\lambda(\boldsymbol{\theta})\right]$$

where $R_\lambda(\cdot)$ is a smooth envelope of $R(\cdot)$

$$R_\lambda(\boldsymbol{\theta}) = \log \int \exp\left(-\frac{1}{2\lambda}\|\boldsymbol{\theta} - \mathbf{v}\|_2^2 - R(\mathbf{v})\right) d\mathbf{v} + \text{const.}$$

Unadjusted Langevin algorithm to target $f_\lambda(\boldsymbol{\theta})$

Using the **Tweedie formula**, the updating rule writes

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \delta\nabla E(\boldsymbol{\theta}^{(t)}) + \frac{\delta}{\lambda}\left(\mathrm{D}_\lambda^*(\boldsymbol{\theta}^{(t)}) - \boldsymbol{\theta}^{(t)}\right) + \sqrt{2\delta}\mathbf{p}^{(t)}$$

$\rightarrow$ in practice, $\mathrm{D}_\lambda^*(\cdot)$ replaced by an off-the-shelf denoiser $\mathrm{D}_\lambda(\cdot)$

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From Bayesian denoising... to plug-and-play denoising

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Split Gibbs sampling (SGS)
Aka consensus Monte Carlo

Objective
One aims at sampling from $f(\boldsymbol{\theta}) \propto \exp\left[-U(\boldsymbol{\theta})\right]$ with $U(\cdot) = E(\cdot) + R(\cdot)$.

Variable splitting
Within a deterministic framework, minimizing $U(\boldsymbol{\theta})$ writes

$$\min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) + R(\boldsymbol{\theta}).$$

An equivalent optimization problem is

$$\min_{\boldsymbol{\theta},\mathbf{z}} E(\boldsymbol{\theta}) + R(\mathbf{z}) \qquad \text{s.t.} \qquad \boldsymbol{\theta} = \mathbf{z}$$

which is relaxed as

$$\min_{\boldsymbol{\theta},\mathbf{z}} E(\boldsymbol{\theta}) + R(\mathbf{z}) + \frac{1}{2\rho^2} \|\boldsymbol{\theta} - \mathbf{z}\|_2^2$$

and iteratively solved by alternate minimization (or ADMM)

- $\boldsymbol{\theta}^{(t)} = \operatorname{argmin}_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) + \frac{1}{2\rho^2} \left\|\boldsymbol{\theta} - \mathbf{z}^{(t-1)}\right\|_2^2$
- $\mathbf{z}^{(t)} = \operatorname{argmin}_{\mathbf{z}} R(\mathbf{z}) + \frac{1}{2\rho^2} \left\|\boldsymbol{\theta}^{(t)} - \mathbf{z}\right\|_2^2$

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Idea: sampling from the augmented distribution

The target distribution $f(\cdot)$ is replaced by the joint distribution

$$f_\rho(\boldsymbol{\theta}, \mathbf{z}) \propto \exp\left[-E(\boldsymbol{\theta}) - R(\mathbf{z}) - \frac{1}{2\rho^2}\|\boldsymbol{\theta} - \mathbf{z}\|_2^2\right]$$

which is sampled using Gibbs steps

- $\boldsymbol{\theta}^{(t)}|\mathbf{z} \sim \exp\left(-E(\boldsymbol{\theta}) - \frac{1}{2\rho^2}\|\boldsymbol{\theta} - \mathbf{z}^{(t-1)}\|_2^2\right)$
- $\mathbf{z}^{(t)}|\boldsymbol{\theta} \sim \exp\left(-R(\mathbf{z}) - \frac{1}{2\rho^2}\|\boldsymbol{\theta}^{(t)} - \mathbf{z}\|_2^2\right)$

Properties

- Defines an asymptotically exact data augmentation:

$$\lim_{\rho \to 0} \left\|f - \tilde{f}_\rho\right\|_{\mathrm{TV}} = 0 \quad \text{with} \quad \tilde{f}_\rho(\theta) = \int f_\rho(\theta, \mathbf{z}) d\mathbf{z}$$

- Simple sampling from each conditional (e.g., MH, E-PO, (MY-)ULA...)
- Can be generalized to handle $f(\boldsymbol{\theta}) \propto \exp[-\sum_{j=1}^J E_j(\boldsymbol{\theta}) - R(\boldsymbol{\theta})]$
- Suitable for distributed sampling (privacy preserving, distributed computing)
- Can be generalized to non-Euclidean splitting (e.g., Bregman divergence).

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Plug-and-play SGS (PnP-SGS)

From Bayesian inversion...

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{b} \quad \text{with} \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \lambda\mathbf{I})$$

Bayesian model

- Likelihood function: $\mathbf{y}|\mathbf{x} \sim f(\mathbf{y}|\mathbf{x}) \propto \exp[-E(\mathbf{x})]$ with $E(\mathbf{x}) = \frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2$
- Prior distribution: $\mathbf{x} \sim f(\mathbf{x}) \propto \exp[-R(\mathbf{x})]$
- Posterior distribution defined as

$$f(\mathbf{x}|\mathbf{y}) \propto \exp\left[-\frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 - R(\mathbf{x})\right]$$

- Augmented posterior distribution defined as

$$f(\mathbf{x}, \mathbf{z}|\mathbf{y}) \propto \exp\left[-\frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 - \frac{1}{2\rho^2}\|\mathbf{x} - \mathbf{z}\|_2^2 - R(\mathbf{z})\right]$$

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  $\rightarrow$ high-dimensional Gaussian sampling (E-PO, RJ-PO,...)
- $\mathbf{z}^{(t)}|\mathbf{x} \sim \exp\left[-\frac{1}{2\rho^2}\|\mathbf{x}^{(t)}-\mathbf{z}\|_2^2 + R(\mathbf{z})\right]$
  $\rightarrow$ Bayesian denoising

Bayesian denoising?

- proximal sampling, e.g., $R(\mathbf{x}) = \|\mathbf{x}\|_1$ (LASSO)
- unadjusted Langevin algorithm, e.g.,
  - regularization-by-denoising: $R(\mathbf{x}) = \mathbf{x}^T(\mathbf{x} - \mathrm{D}(\mathbf{x}))$
  - convex ridge regularization: $\nabla R(\mathbf{x}) = \mathbf{W}^T\mathrm{act}(\mathbf{W}\mathbf{x})$
  - normalizing flow
  - score-based generative model: $\nabla R(\mathbf{x}) = \mathrm{s}(\mathbf{x})$]
- plug-and-play sampling: *denoising diffusion probabilistic model* (DDPM)

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Outline

Conclusion
Monte Carlo methods

What?

- non-iterative or iterative algorithms to sample from a target distribution

How?

- resorting to (pseudo-) random generators embedded in a computer,
- designing elementary algorithmic building blocks,
- combining the building blocks to implement efficient algorithms

Why?

- having a set of samples distributed according to the target posterior distribution,
- approximating quantities of interest by Monte Carlo integration

Conclusion
Pros and *cons*

Major assets

- high flexibility of the models (likelihood functions and prior distributions)
  (non-convex or non-smooth potentials, mixture models, discrete and/or continuous parameters...)
- genericity and simplicity of implementation
  (with "a few" calculations)
- point estimation: several possible choices (beyond MMSE and MAP estimators)
- comprehensive/full description of the target posterior distribution: credibility intervals!

Major challenge: computational complexity
In high dimension, in case of naive implementation

- potentially expensive generation of a sample $\theta^{(t)}$
- need for numerous samples $\theta^{(1)}, \ldots, \theta^{(T)}$
- potentially slow convergence

$\rightarrow$ ***computational times*** *and* ***memory!***

Conclusion
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Conclusion
Some opportunities

Recent advances

- algorithms inspired by models brought by statistical physics
  Langevin diffusion, Hamiltonian Monte Carlo,...
- algorithms inspired by optimization-based methods
  proximal Monte Carlo, split Gibbs sampling...
- interplay with advanced machine learning models, tools and concepts
  deep generative models, optimal transport

Expected gains

- algorithm efficiency
  $\rightarrow$ *reducing the computational burden*
  $\rightarrow$ *solving a high dimensional problems*
- approximation guarantees
  $\rightarrow$ *asymptotic and non-asymptotic bounds*
- data-driven models
  $\rightarrow$ *implicit definition of the models*
  $\rightarrow$ *emulations of even more expensive physical models*

Conclusion

Not discussed today

- *Approximate Bayesian Computation* (ABC)
- Quasi-Monte Carlo methods
- Monte Carlo drop-out
- Bayesian variational methods
- Sequential Monte Carlo methods (aka particle filtering)
- sampling from (deep) generative models
- machine learning-based improvements of Monte Carlo methods
- ...

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Bayesian methods for uncertainty quantification

Part II – Monte Carlo statistical methods

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